

# MORPHOLOGICAL OPERATIONS

## 20XW91 - Computer Vision

# MATHEMATICAL MORPHOLOGY

### Definition

Mathematical morphology is a powerful image–processing technique used to extract and analyze image components that are useful for representation and description.

### Mathematical Foundation

Mathematical morphology is based entirely on Set Theory.

### Main Motivation

The primary goal is to obtain and analyze the structural information present in an image.

### Image Representation

In morphology, sets represent objects in an image.

# STRUCTURING ELEMENT

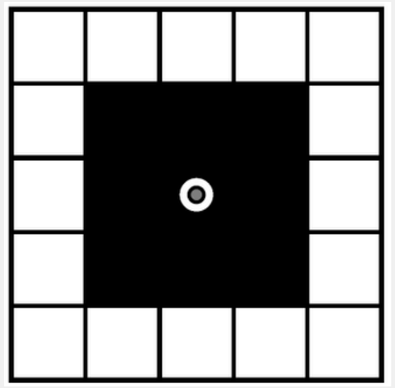
**Mathematical morphology is a collection of non-linear processes which can be applied to an image to remove details smaller than a certain reference shape, which is called structuring element.**

**Similarity to Convolution**

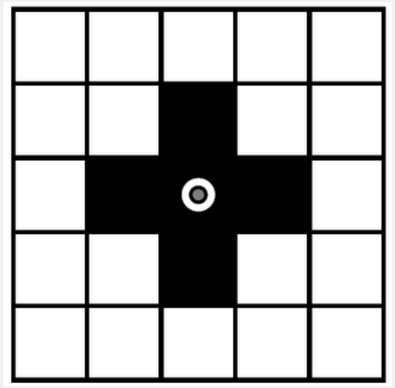
Morphological processing is similar to the convolution process:

- Convolution uses a **kernel** and **arithmetic operations**.
- Morphology uses a **structuring element** and **logical/set operations**.

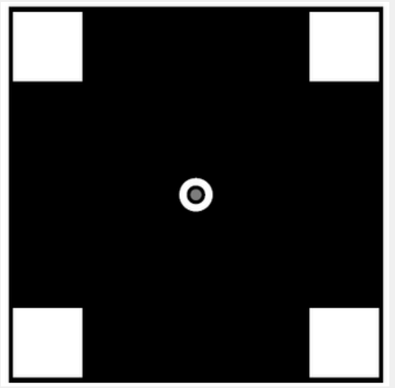
- **The structuring element in a morphological operation plays an important role with its different shape and size.**
- **Shape and size are defined by a number of 0s and 1s in the structuring elements**



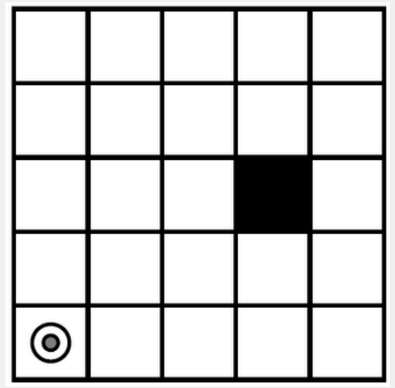
N8 (8 – Neighbourhood centred)



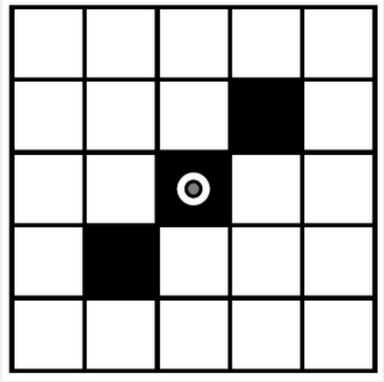
N4 (4 – Neighbourhood centred)



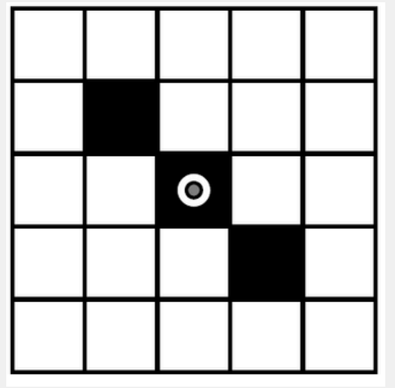
Flat plus



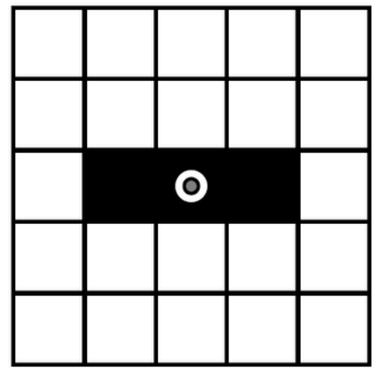
Shifted version



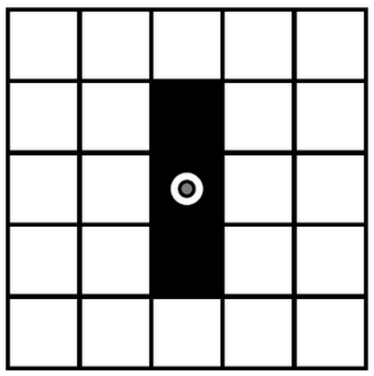
line-structuring element of 45°



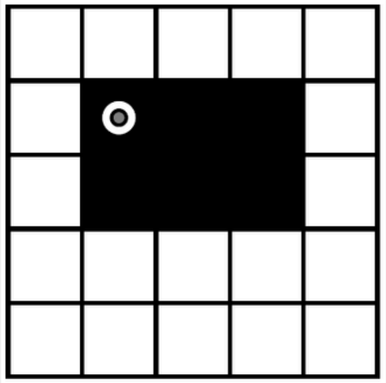
line-structuring element of 135°



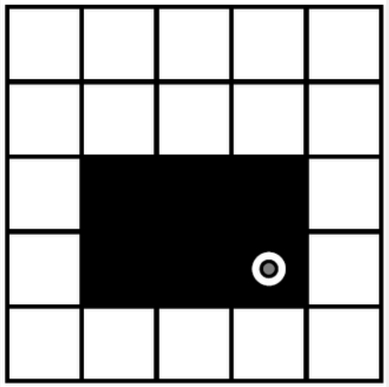
Horizontal structuring element
with size 1 × 3



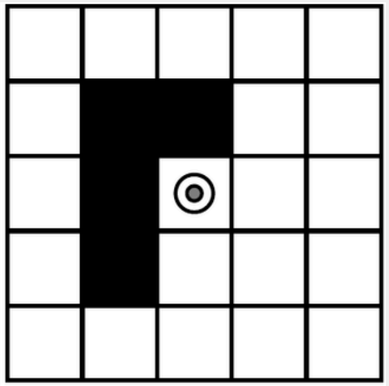
Vertical structuring element
with size 3 × 1



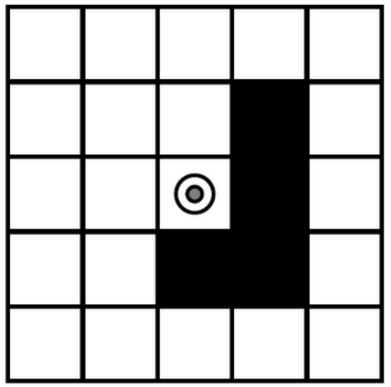
2 × 3 sized rectangular



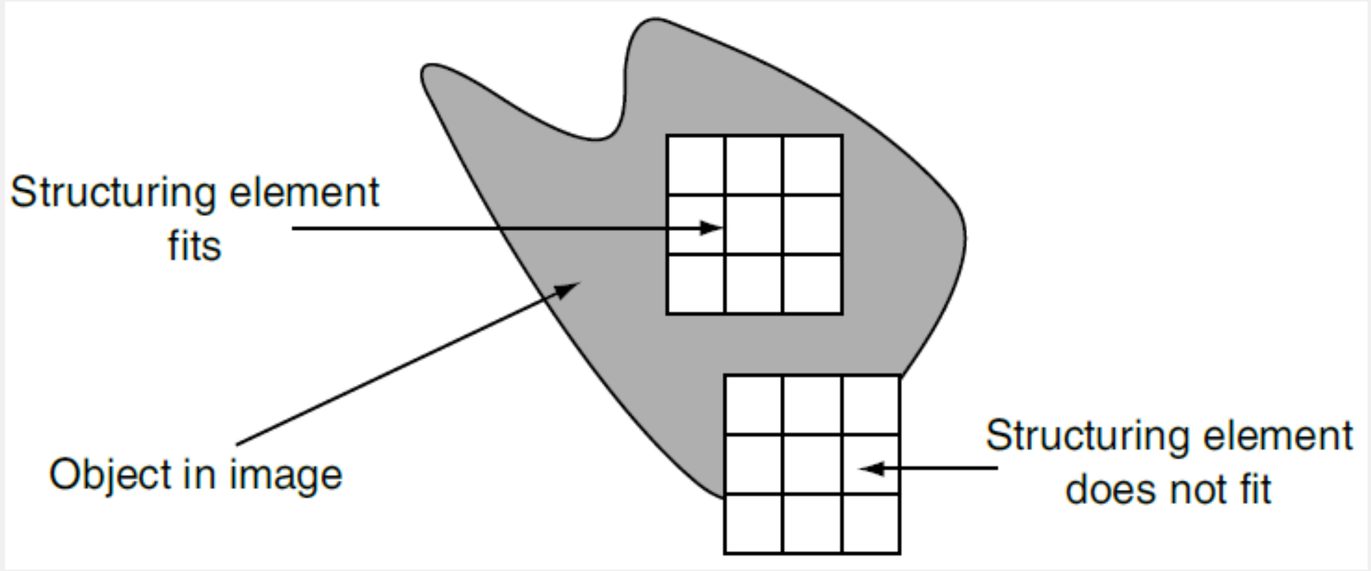
Reflected
structuring element



‘L’ Shaped structuring element



Reflected version of
structuring element




## Basic Principle

Morphological operations are performed by moving a structuring element (SE) across the binary image.

## Logical Operations

At each position, a logical operation is carried out between the structuring element and the pixels covered by it.

## Structuring Element Positioning

The structuring element is centered on every pixel of the image, one position at a time.

## Output Generation

The result of the logical operation at each location is stored in the corresponding pixel position of the output image.

# EROSION

Erosion is the counter–process of dilation. If dilation enlarges an image then erosion shrinks the image.

Erosion operator takes two inputs,

- input image
- structuring element

The way the image is shrunk is determined by the structuring element.

# EROSION

1

The structuring element is normally smaller than the image with a 3 × 3 size.

2

This will ensure faster computation time when compared to larger structuring-element size.

3

The erosion process will move the structuring element from left to right and top to bottom.

4

At the centre position, indicated by the centre of the structuring element, the process will look for whether there is a complete overlap with the structuring element or not.

5

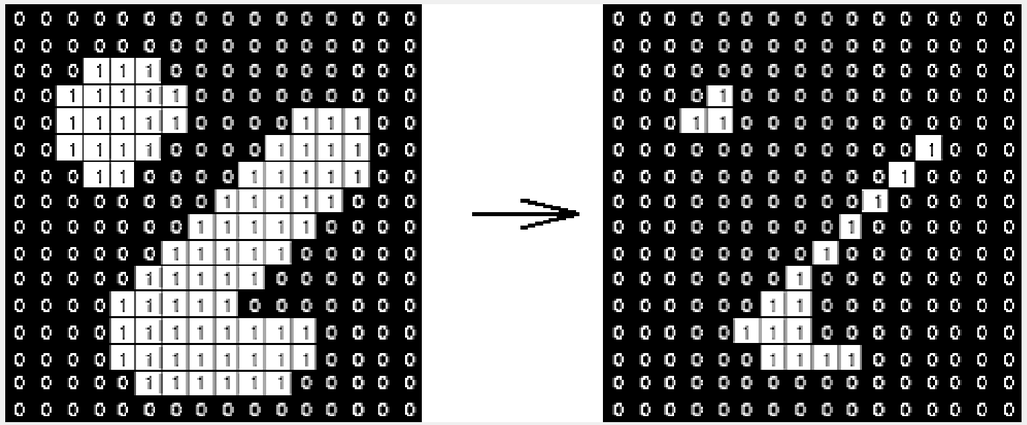
If there is no complete overlapping, then the centre pixel indicated by the centre of the structuring element will be set white or 0.

# KEY POINTS

- Dual of dilation
- Shrinks the object in the input image
- Size of input and output image is same
- Nature of thinning(shrinking) is determined by the structuring element.

## Erosion

# Morphological Operations



**Erosion of Binary Image – In a binary image, a pixel is set to 1 if all of the neighbouring pixels have the value 1.**



**Erosion of Grayscale Image – The value of the output pixel is the minimum value of all pixels in the neighbourhood.**

# Erosion

- The outcome element z is considered only when the structuring element is a subset or equal to the binary image X.

## Equation

Let us define X as the reference binary image and B as the structuring element. Erosion is defined by the equation

$$X \ominus B = \left\{ z \middle| (B)_z \subseteq X \right\}$$

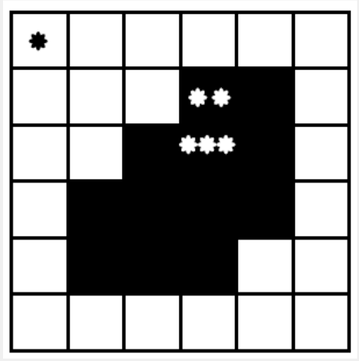
## Erosion

# Morphological Operations

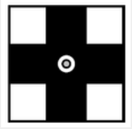
- The erosion process starts at position *.
- Here, there is no complete overlapping, and so the pixel at the position * will remain white.

- The structuring element is then shifted to the right and the same condition is observed.
- At position **, complete overlapping is not present; thus, the black square marked with ** will be turned to white.

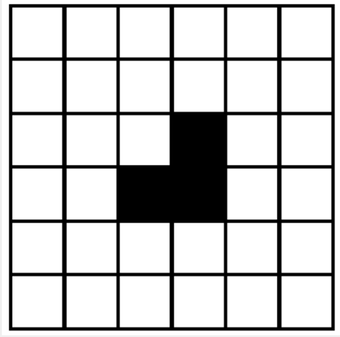
- The structuring element is then shifted further until its centre reaches the position marked by ***.
- Here, we see that the overlapping is complete, that is, all the black squares in the structuring element overlap with the black squares in the image.
- Hence, the centre of the structuring element corresponding to the image will be black.



Input Image



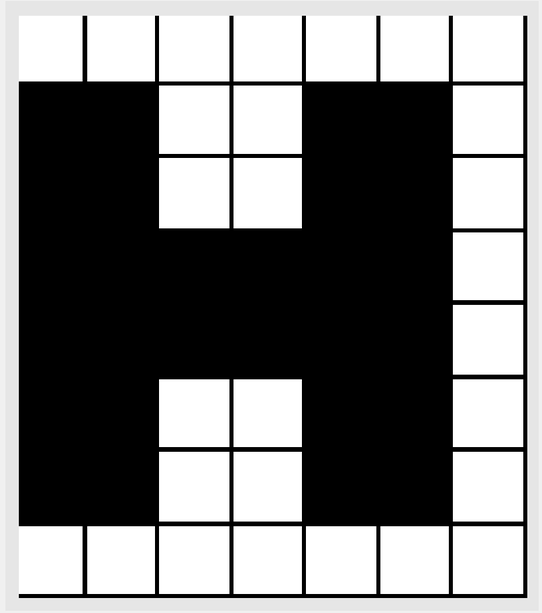
Structuring Element



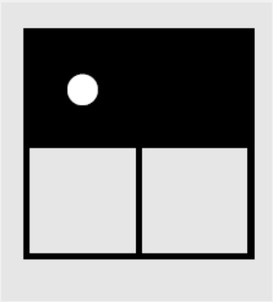
Output Image

The input image and structuring element are given below. Find the eroded version of the input image.

## Input image :



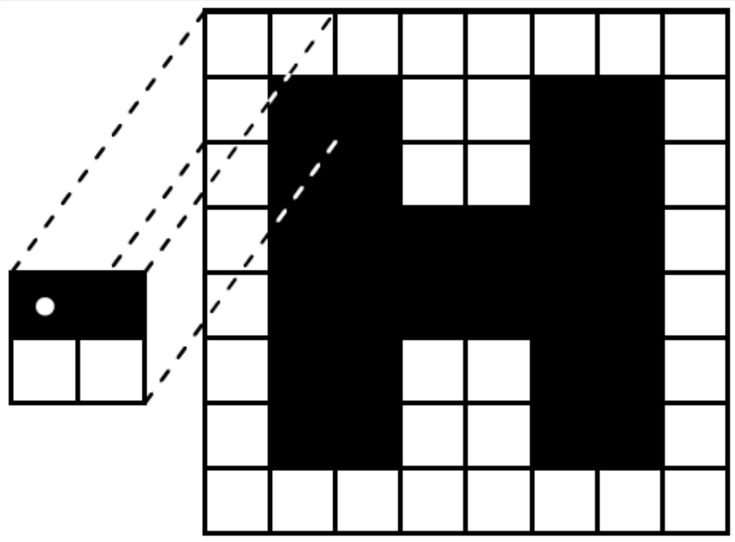
## Structuring Element :



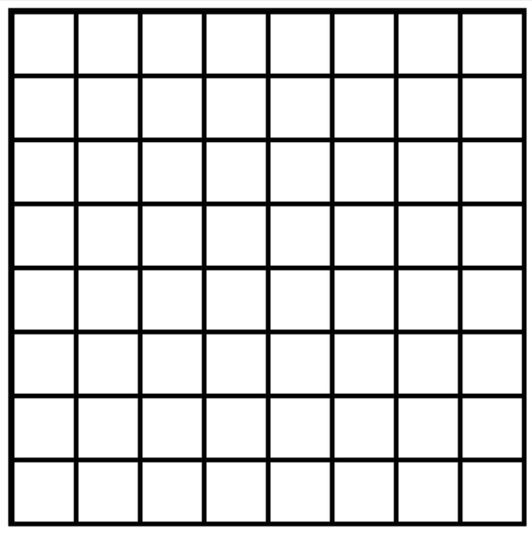
## STEP 1 :

Compare the structuring element with the image. If all pixels overlap, the output pixel is ON; otherwise, it is OFF. In the first row, there is no complete overlap, so all pixels are OFF.

### Input image :



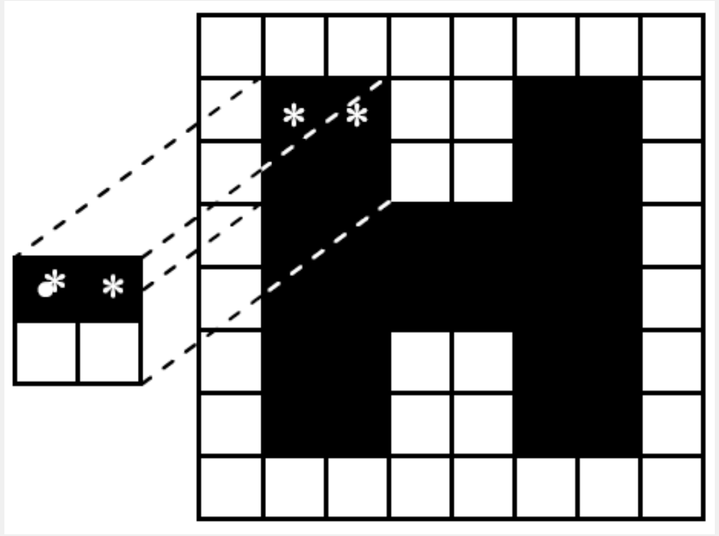
### Output image :



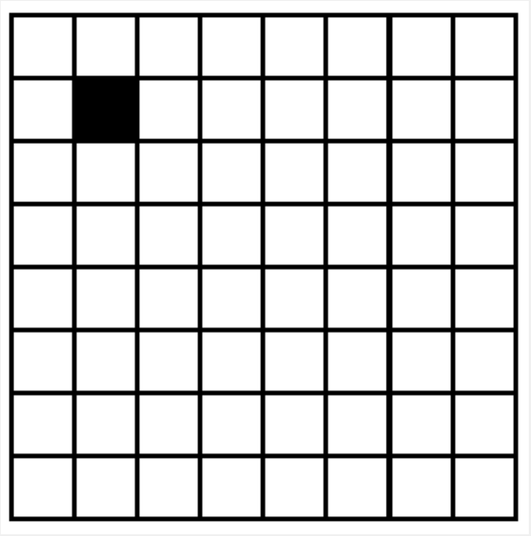
## STEP 2 :

The structuring element is mapped to the second row. If there is no match, shift it right until a perfect match is found, then turn the output pixel ON.

### Input image :



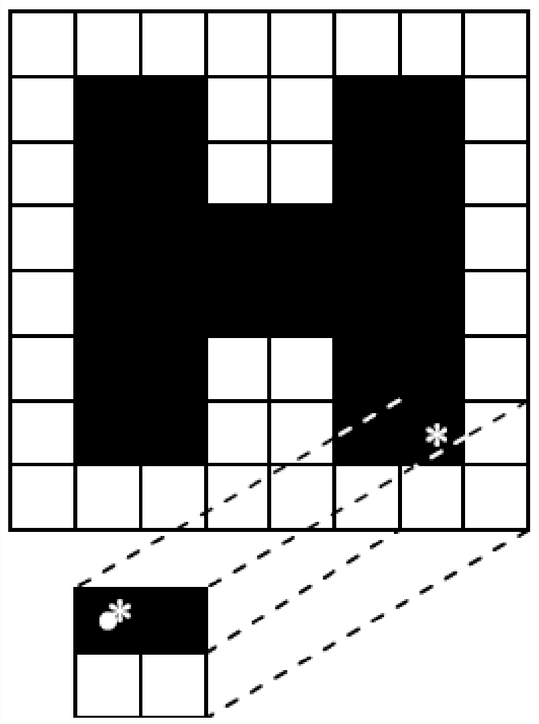
### Output image :



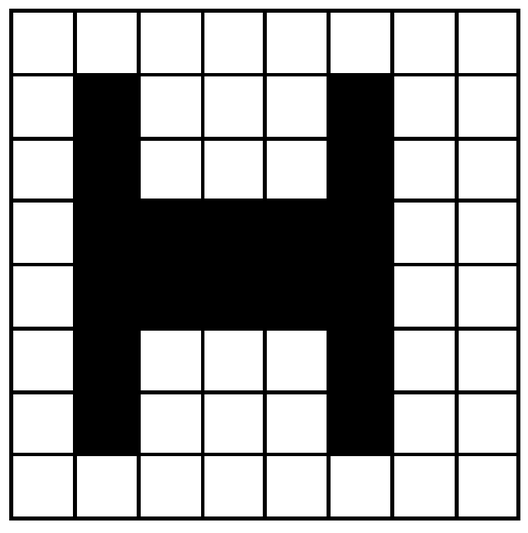
## STEP 3 :

Repeat the process for all pixels. The resulting image is the eroded image.

### Input image :



### Output image :



# APPLICATIONS

## Counting objects

The erosion process can be used to disconnect connected objects so that they can be separated and counted using labelling algorithms.

## Removing Salt Noise

The erosion process can be used to remove salt noise(white dots) from an image. This affects the quality of the image, as a result opening process(erosion followed by dilation) is generally used for this purpose.

## Feature Thinning

Shrinks the edges of shapes to find exact object boundaries or prepare images for machine learning.

# DILATION

Dilation is one of the most common morphological operation along with Erosion. Dilation is used to add pixels to the boundary of the input image making the object more visible and fill small voids in the image.

Dilation operator takes two inputs,

- input image
- structuring element

The structuring element determines the effect of dilation on the input image.

# DILATION

**1**

This structuring element is smaller in size compared to the image itself, and normally the size used for the structuring element is 3 × 3.

**2**

The dilation process is similar to the convolution process, that is, the structuring element is reflected and shifted from left to right and from top to bottom, at each shift

**3**

The process will look for any overlapping similar pixels between the structuring element and that of the binary image.

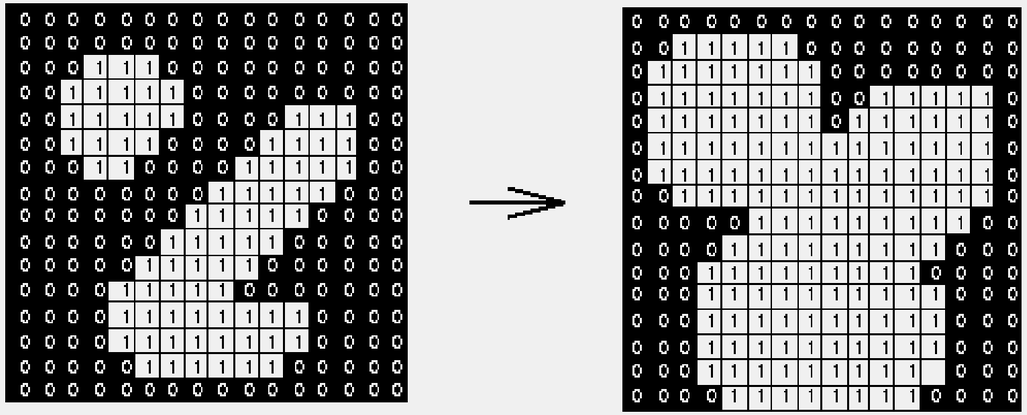
**4**

If there exists an overlapping then the pixels under the centre position of the structuring element will be turned to 1 or black.

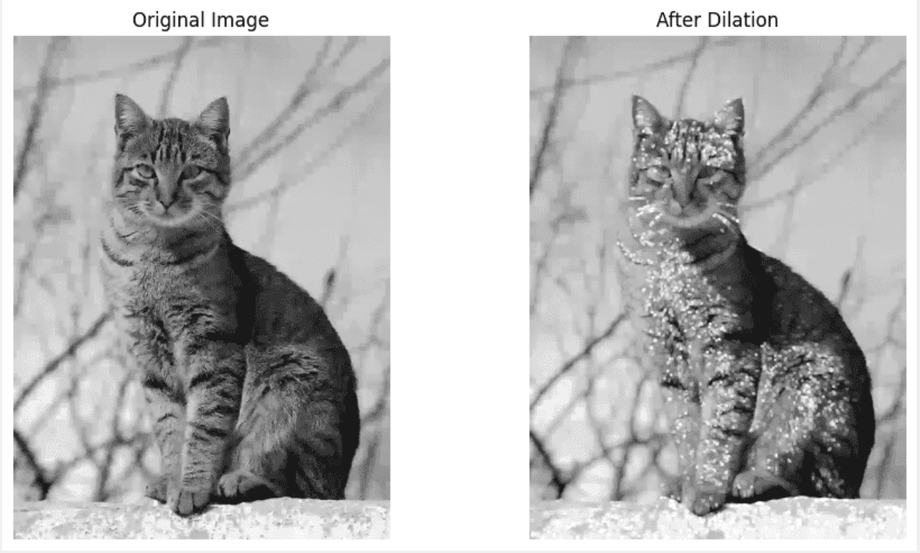
# KEY POINTS

- Dual of Erosion
- Grows the object and fill any voids in the input image
- Size of input and output image is same
- Nature of thickening is determined by the structuring element

## Dilation



**Dilation of Binary Image – In a binary image, a pixel is set to 1 if any of the neighbouring pixels have the value 1.**




**Dilation of Grayscale Image – The value of the output pixel is the maximum value of all pixels in the neighbourhood.**

# Dilation

- When the image X is dilated by the structuring element B, the outcome element z would be that there will be at least one element in B that intersects with an element in X.
- If this is the case, the position where the structuring element is being centred on the image will be ‘ON’.

## Equation

Let us define X as the reference image and B as the structuring element. The dilation operation is defined
by Eq.

$$X \oplus B = \left\{ z \middle| \left[ \left( \hat{B} \right)_z \cap X \right] \subseteq X \right\}$$

where ˆB is the image B rotated about the origin.

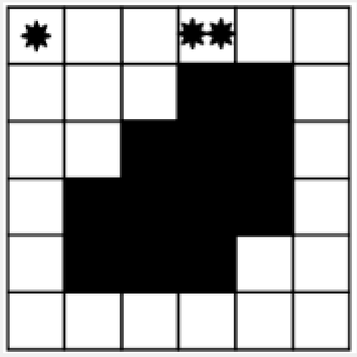
## Dilation

# Morphological Operations

- Initially, the centre of the structuring element is aligned at position *.
- At this point, there is no overlapping between the black squares of B and the black squares of X; hence at position * the square will remain white

- This structuring element will then be shifted towards right.
- At position **, we find that one of the black squares of B is overlapping or intersecting with the black square of X.
- Thus, at position ** the square will be changed to black.

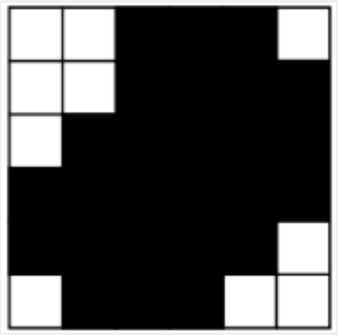
- Similarly, the structuring element B is shifted from left to right and from top to bottom on the image X to yield the dilated image



Input Image



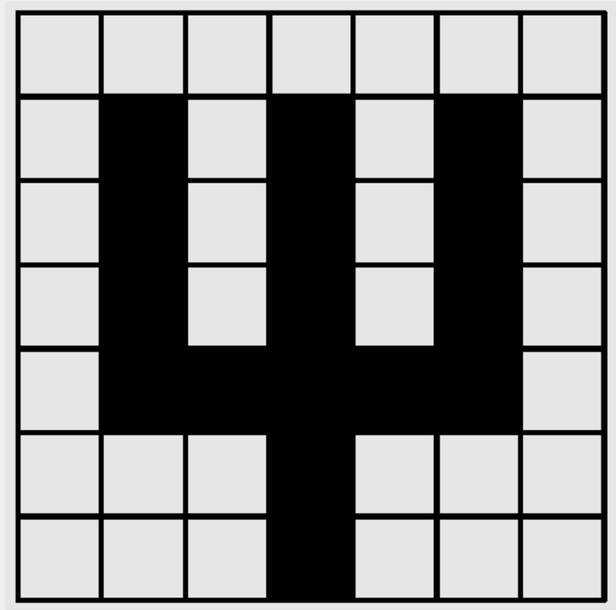
Structuring Element



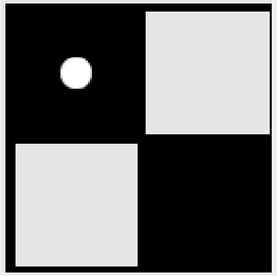
Dilated Image

Using the input image and structuring element as given below, find the dilated version of the input image.

## Input image :



## Structuring Element :

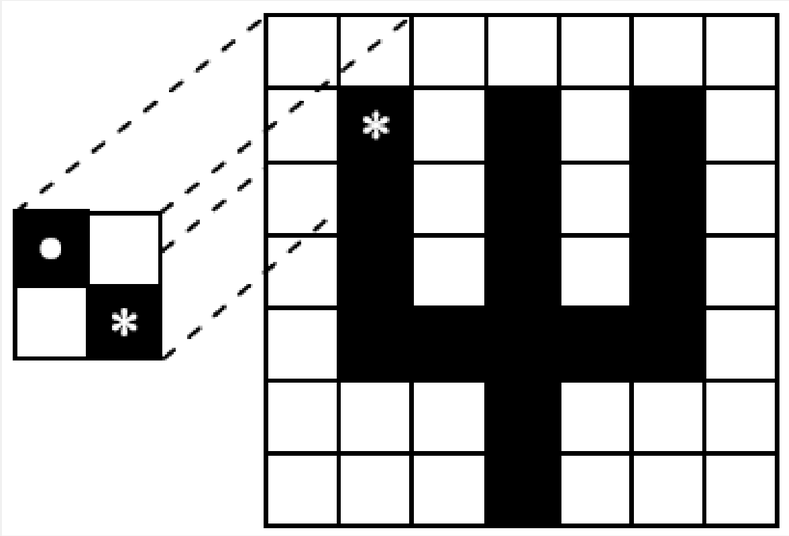


## Dilation

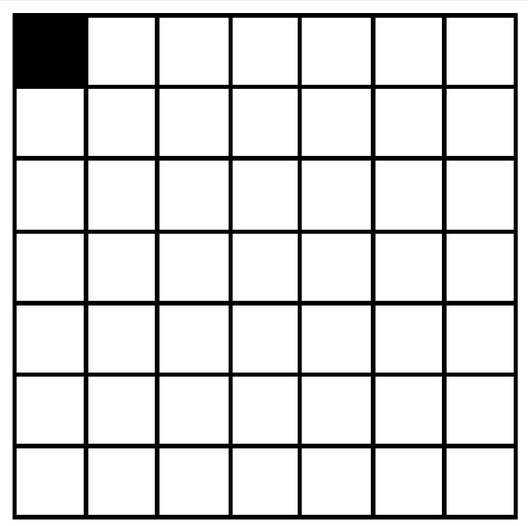
### STEP 1 :

Place the structuring element at the centre pixel. If there is overlap, turn the corresponding output pixel ON.

### Input image :



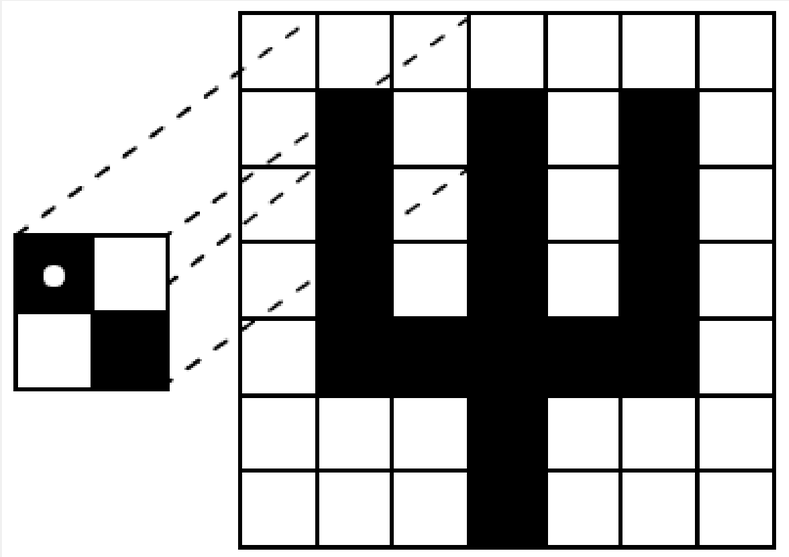
### Output image :



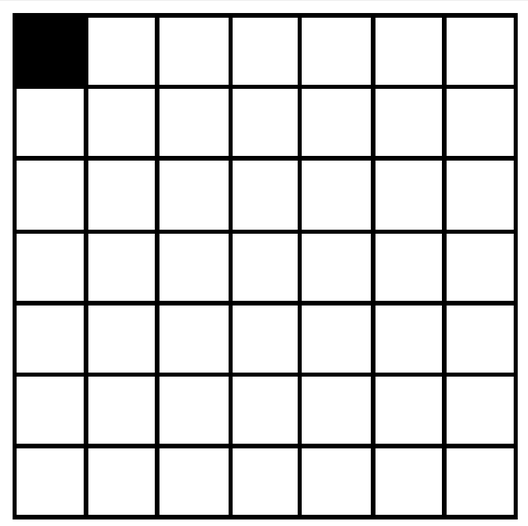
## STEP 2 :

Shift the structuring element from left to right. If there is no overlap, the output remains OFF.

### Input image :



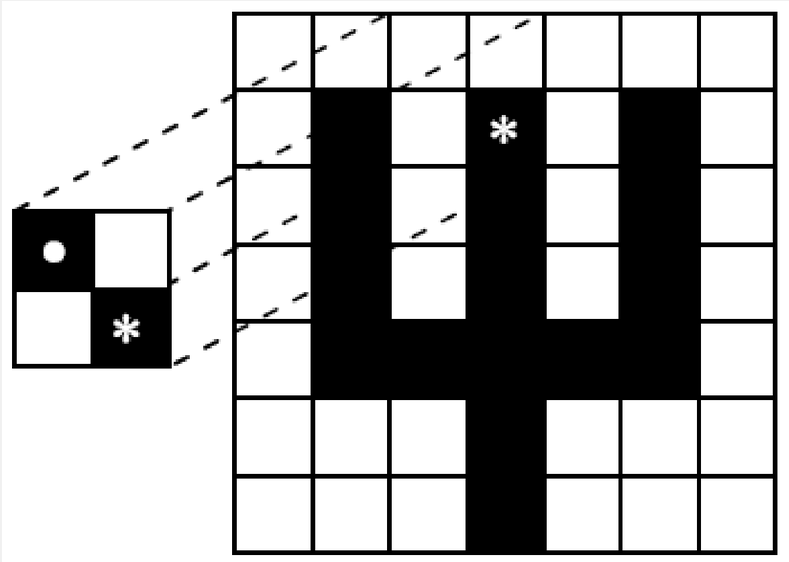
### Output image :



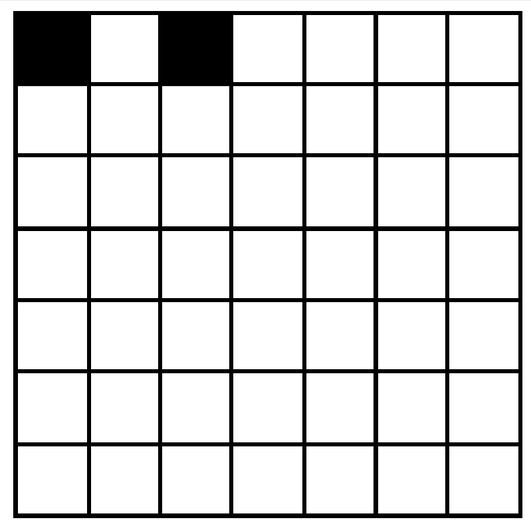
## STEP 3 :

Continue shifting right. Whenever overlap occurs, turn the corresponding output pixel ON.

### Input image :



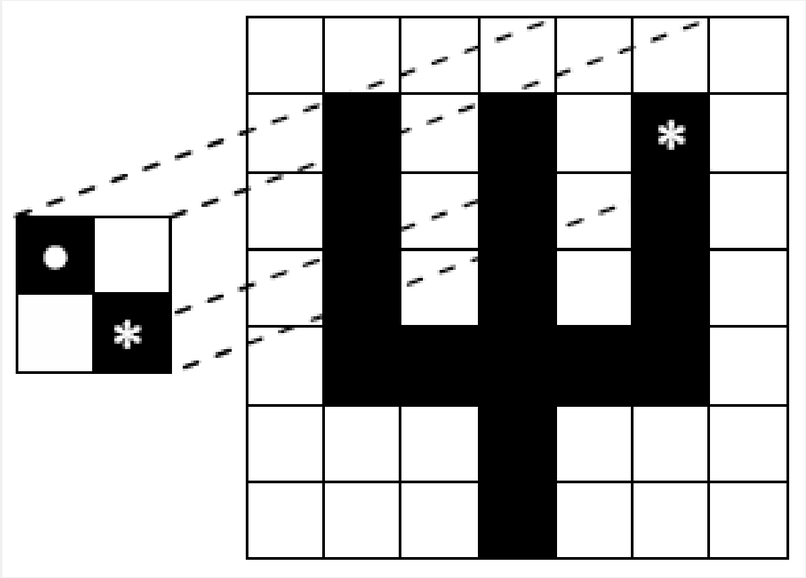
### Output image :



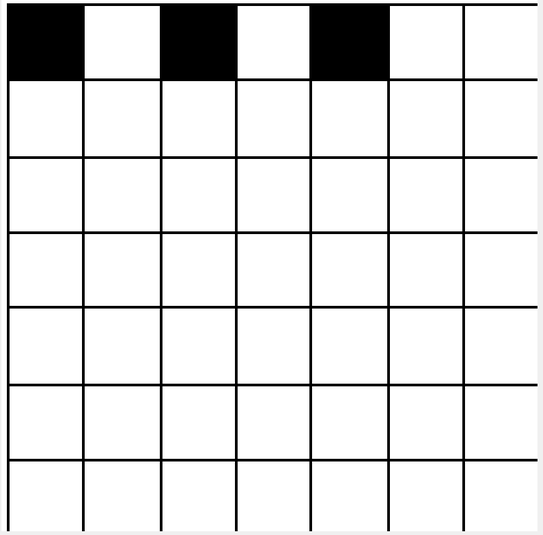
## STEP 4 :

**Repeat the shifting process and turn ON the output pixels wherever overlap occurs.**

### Input image :



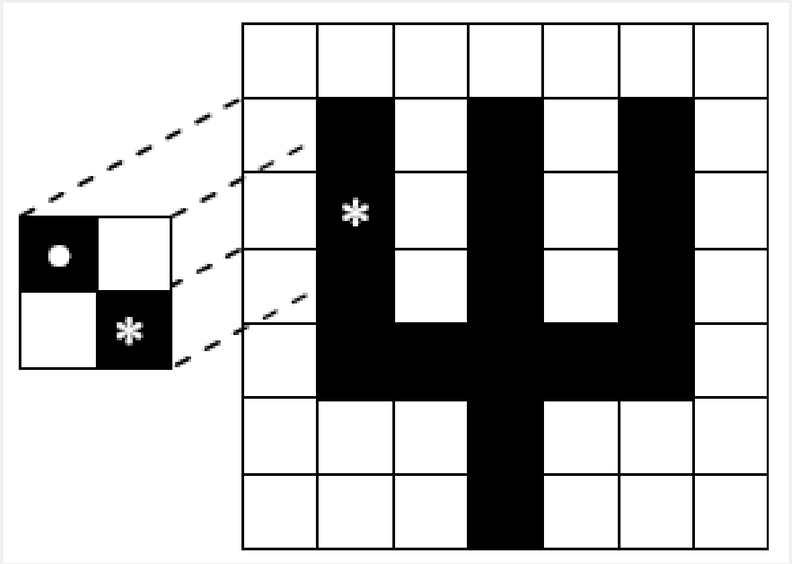
### Output image :



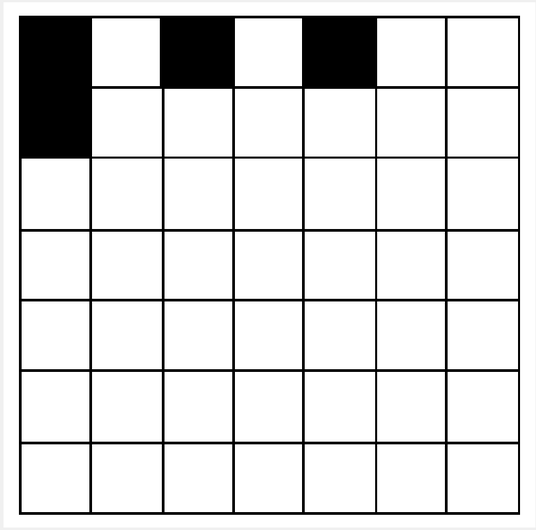
## STEP 5 :

When there is no overlap or the boundary is reached, move the structuring element to the next row.

### Input image :



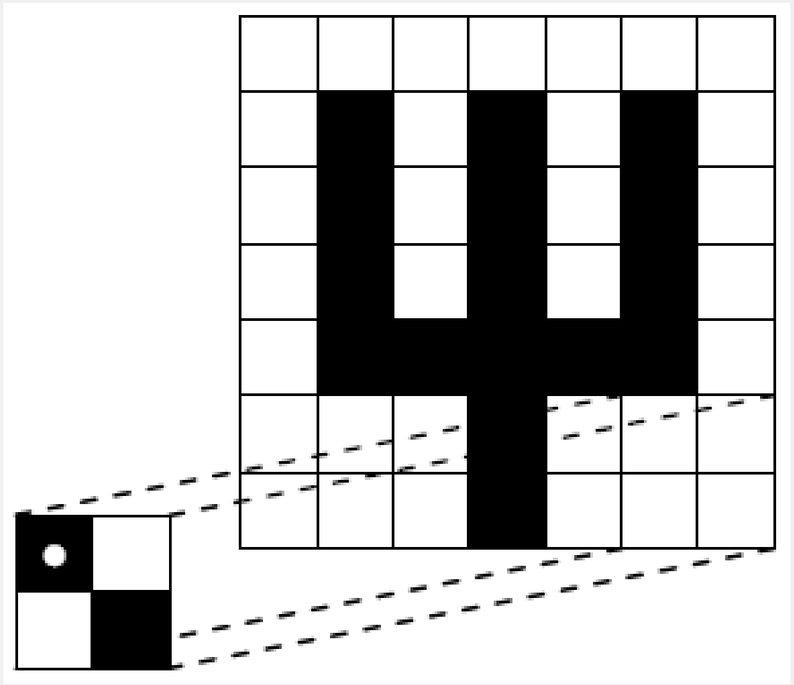
### Output image :



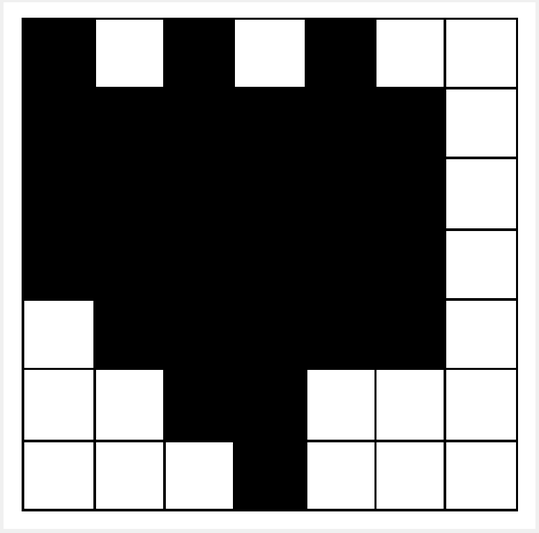
## STEP 6 :

Repeat the process for the entire image to obtain the final dilated image.

### Input image :



### Output image :



# APPLICATIONS

## Removing Pepper Noise

The dilation process is perfect for filling the small voids(pepper noise) in images.

## Edge Detection

The dilation process can be used to detect edges in the input image by performing the dilation on the input image and subtracting the original image from the dilated image. This leaves behind the boundary created by extra foreground pixels as a result of dilation process.

## Filling Holes

Closes small dark cavities, internal voids, or missing pixels inside an object.

## Morphological Operations



Input image



(a) Dilated image



(b) Eroded image

# OPENING

**Erosion followed by Dilation**

$$X \circ Y = (X \ominus Y) \oplus Y$$

**Removes small objects from the foreground (usually taken as the bright pixels) of an image, placing them in the background**

**The opening is formed from the portions of the object where the structuring element can fit, followed by expansion of those surviving portions.**

# What does opening actually do ?

## smooths the contour of an object

- Tiny protrusions and irregularities may not survive erosion.
- After dilation, the larger surviving object is restored, but those tiny irregularities are gone.
- So the contour becomes smoother.

## breaks narrow isthmuses

- An isthmus is a narrow connection between two larger parts.
- If the connecting portion is narrower than the structuring element, erosion removes it.
- After dilation, the two large portions remain, but the narrow connection does not necessarily return.
- Therefore opening can separate objects connected by narrow bridges.

## eliminates thin protrusions

- If the protrusion is too thin for the structuring element to fit inside it, erosion removes it.
- The later dilation restores the main body but not the eliminated protrusion.
- So opening is particularly useful for removing thin unwanted structures.

## Opening

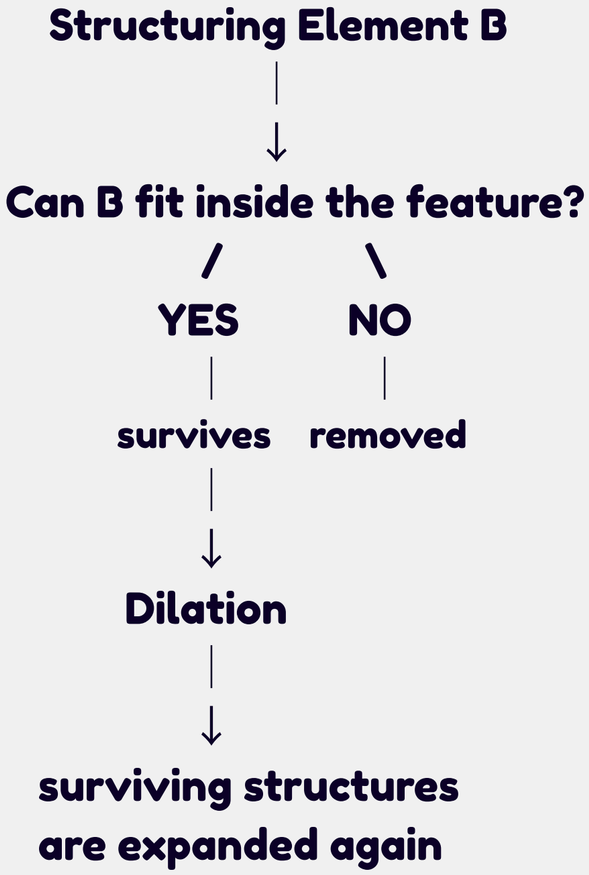
## Morphological Operations

Erosion generally:
shrinks objects
removes small foreground components
removes thin portions
breaks narrow connections
can enlarge dark/background regions

**Step 1: Erosion**

Dilation expands the remaining structures.
The important thing is that dilation operates on the eroded result, not on the original image.
This is why opening can remove small objects while retaining the general shape of larger objects.

**Step 2: Dilation**



# APPLICATIONS

## Separating shapes

01

Opening process can be used for separating particular shaped objects. The opening is performed in the input image(left) containing line and circles to get the circles in the output image(right) with the help of a disk shaped structuring element.

## Removing Salt Noise

02

The erosion process can be used to remove salt noise(white dots) from an image. This affects the quality of the image, as a result opening process(erosion followed by dilation) is generally used for this purpose.

# CLOSING

**Dilation followed by Erosion**

$$X \cdot Y = (X \oplus Y) \ominus Y$$

**Removes small holes in the foreground, changing small islands of background into foreground.**

**The closing is formed by first expanding the object using dilation and then shrinking it using erosion.**

# What does closing actually do ?

## Smooths object boundaries

- Small indentations, cracks, and irregularities along object boundaries are filled during dilation.
- After erosion, the overall shape is restored, but the boundary becomes smoother.

## Connects narrow gaps

- If two objects are separated by a small gap that is narrower than the structuring element, dilation connects them.
- The subsequent erosion preserves this connection.

## Eliminates small dark regions

- Tiny background regions that are smaller than the structuring element are removed during dilation and do not reappear after erosion.

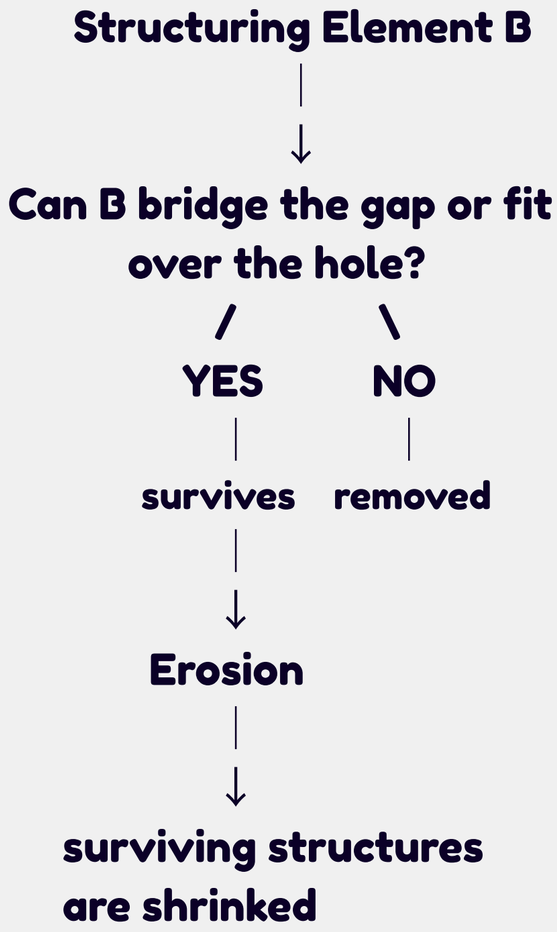
## Closing

Dilation generally:
Expands objects
Fills small holes and gaps
Connects nearby objects
Reduces small dark regions
Enlarges foreground areas

**Step 1: Dilation**

Erosion shrinks the dilated result and restores the object's approximate original size.
The important point is that erosion operates on the dilated image, not on the original image.
This is why closing can fill holes and connect nearby objects while retaining the overall shape of larger objects.

**Step 2: Erosion**



# APPLICATIONS

## Connecting Nearby Objects

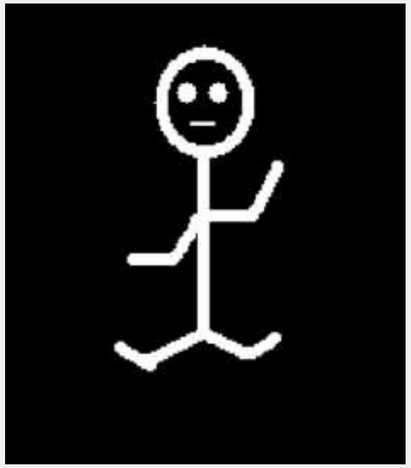
01

Closing can connect objects that are separated by small gaps. If the gap is smaller than the structuring element, dilation joins the objects and erosion preserves the connection.

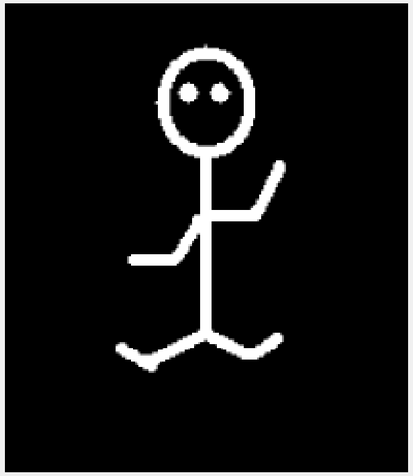
## Removing Pepper Noise

02

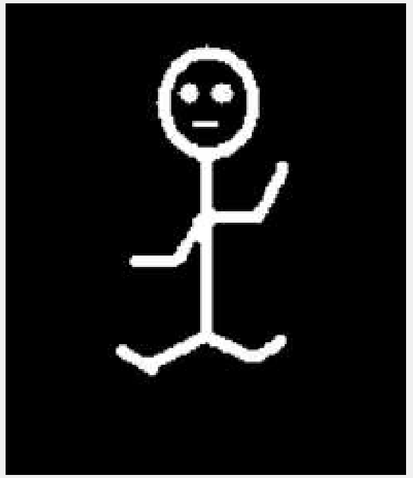
Closing can be used to remove small black dots or pepper noise from an image. Dilation fills the small dark regions, and the following erosion restores the overall size of the foreground objects.



Input image



(a) Opened image



(b) Closed image

# MORPHOLOGICAL ALGORITHMS

# Hit or Miss Transform

- **Purpose:** Used for template matching in binary images.
- It uses two disjoint template sets:
  - **B** → matches the foreground of the image.
  - **(W − B)** → matches the background of the image.
- The two template sets do not overlap.
- The transformation is obtained by taking the intersection of two erosions:
  - Erosion of the foreground image X with B.
  - Erosion of the background image $X^c$ with (W − B).

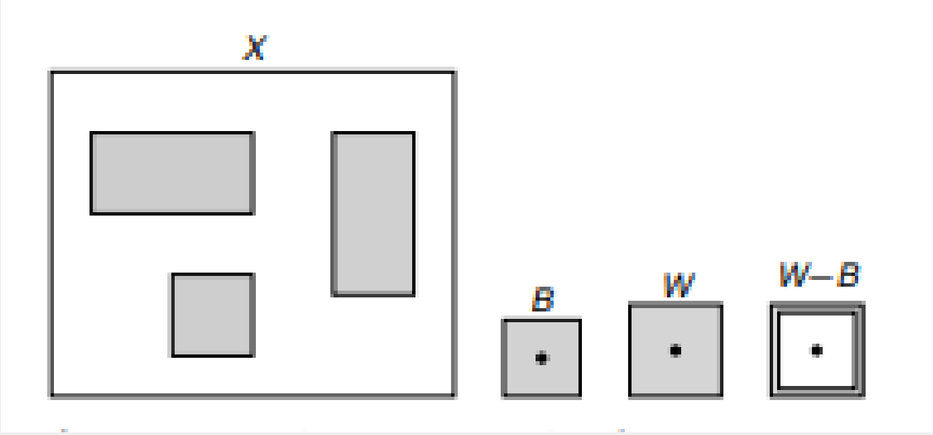
# Hit or Miss Transform

$$HM(X,B) = (X \ominus B) \cap (X^c \ominus (W - B))$$

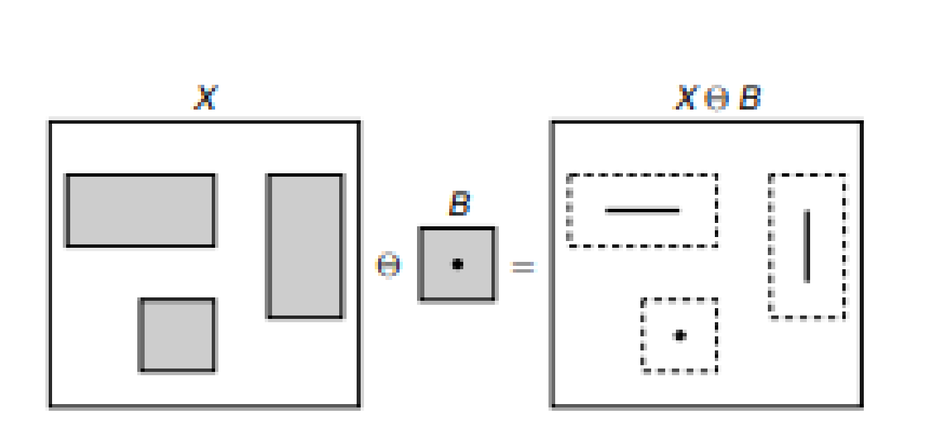
- $X^c$ represents the **complement/background** of the input image.
- **W** represents the **small window** containing the templates.
- The result identifies locations where **both foreground** and **background patterns match simultaneously**.
- Therefore, the Hit-or-Miss Transform is useful for detecting a **specific shape or pattern** in a binary image.

# Image example:





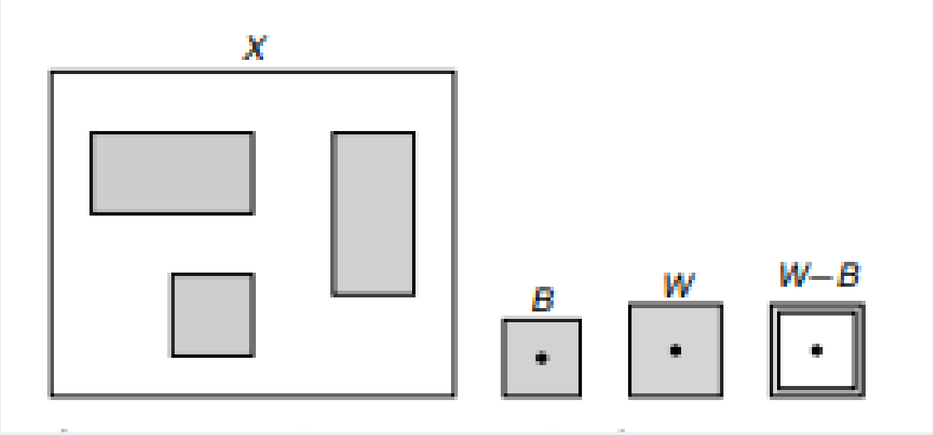

Input image X, structuring element B, W, W−B



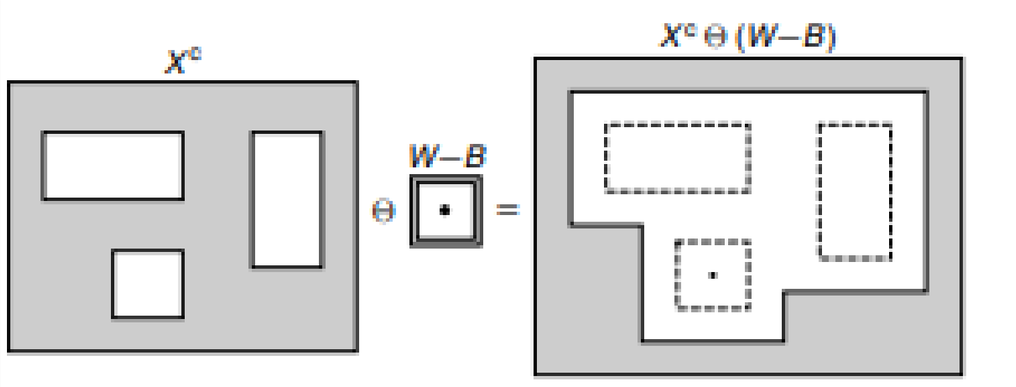

Eroded image of X

# Image example:





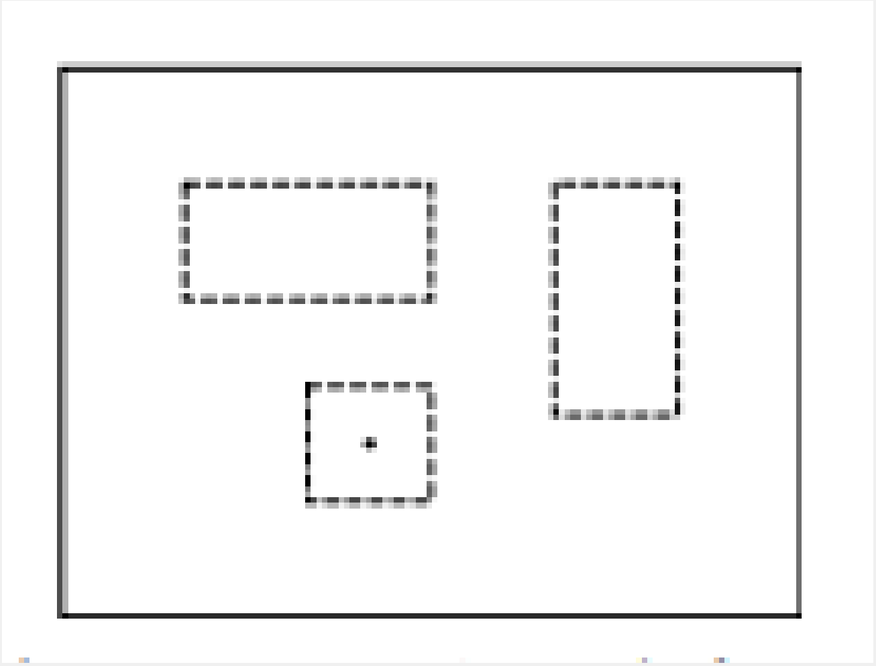

Input image X, structuring element B, W, W–B




Eroded image of $X^c$

# Image example:



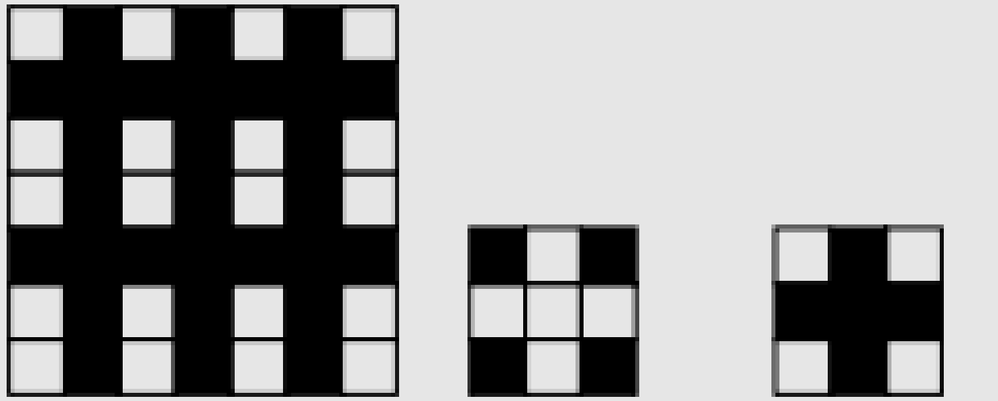


**Intersection result of**
**Eroded image of X and Eroded image of $X^c$**

# Question



**Example 10.4** *The input image and structuring elements are shown below. Find the hit or miss transformation for the input image.*

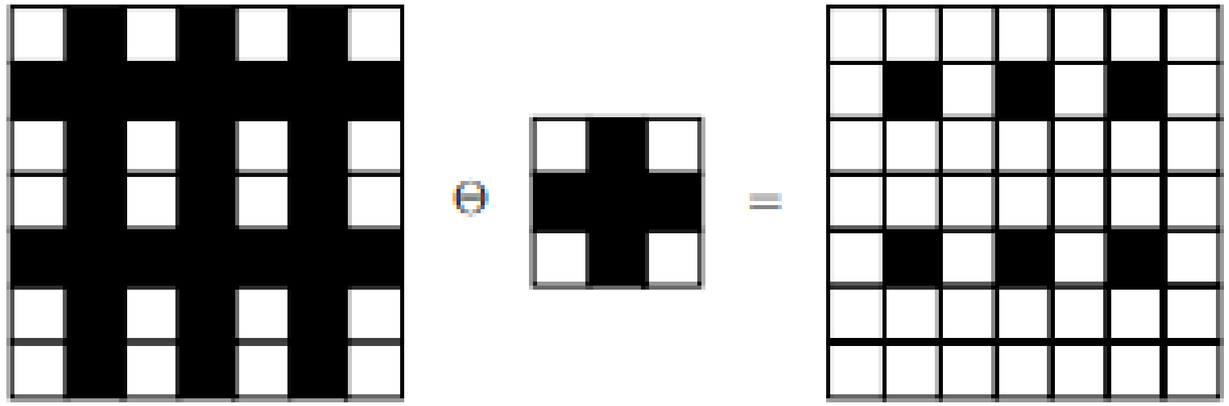


(a) Input image (b) Structuring element $B$ (c) Structuring element $W-B$

# Solution



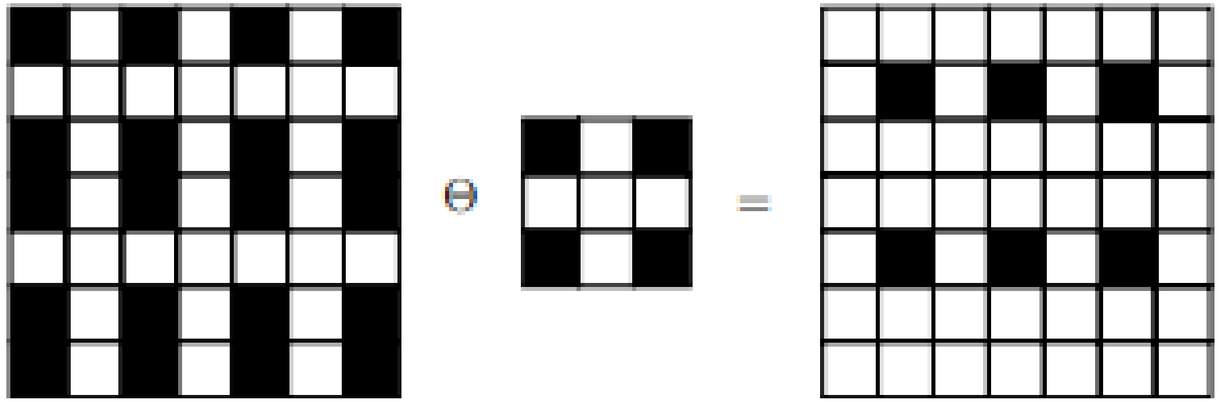
***Step 1*** From the definition of hit-or-miss transformation, we have to find the erosion of the input image with the structuring element $B$. The result of the erosion operation is shown below.



# Solution



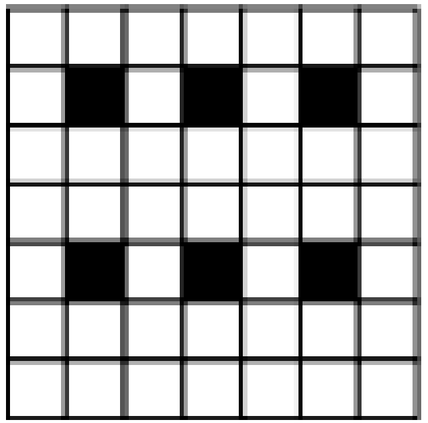
***Step 2*** Next, find the erosion of the complement of the input image with the structuring element $W-B$; we get the output image as shown below.



# Solution



***Step 3*** From the result of steps 1 and 2, the hit-or-miss transformed input image is found from the intersection of the result of the image in steps 1 and 2. The resultant image is shown below.



# Boundary Detection

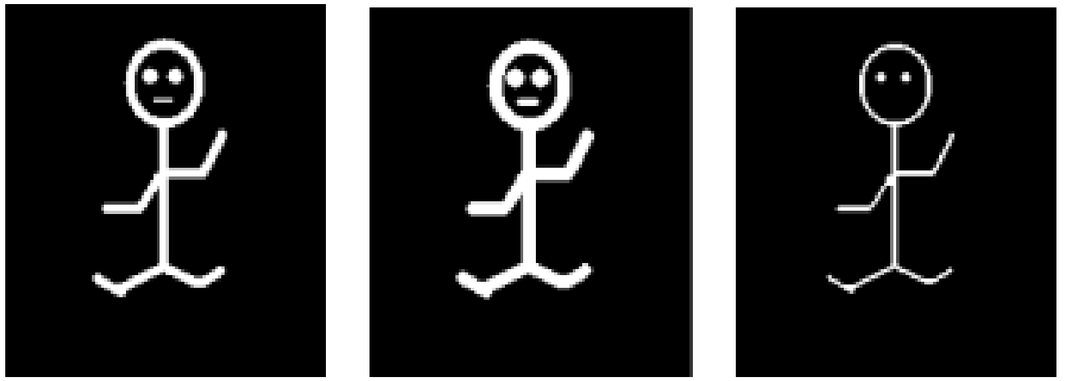
- Morphological operations can be effectively used to **detect boundaries** in a binary image **X.**
- The boundary image is represented by **Y.**
- Two common morphological approaches are used for boundary detection:
  - **$Y = X - (X \ominus B)$ -> Inner boundary**
  - **$Y = (X \oplus B) - X$ -> Outer boundary**
- **$\ominus$ (Erosion):** Removes pixels from the boundaries of the foreground.
- **$\oplus$ (Dilation):** Expands the foreground region by adding pixels around its boundary.
- **$-$ (Set-theoretical subtraction):** Extracts the difference between two binary images.

# Boundary Detection

- **Erosion-based boundary detection:**
  - Erode the original image using structuring element **B**.
  - Subtract the eroded image from the original image.
  - The remaining pixels represent the **inner boundary**.
- **Dilation-based boundary detection:**
  - Dilate the original image using **B**.
  - Subtract the original image from the dilated image.
  - The added pixels represent the **outer boundary**.
- These techniques can be used to **extract the boundary of objects** in binary image processing.

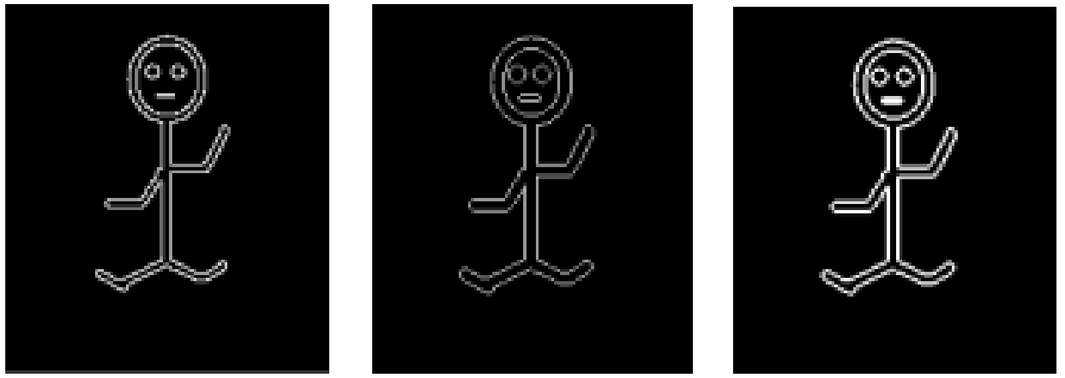
# Image example:





(a) Original image $X$ (b) Dilated image (c) Eroded image

**Fig. 10.43** *Dilated and eroded image of X*



(a) $X - (X \ominus B)$ (b) $(X \oplus B) - X$ (c) $(X \oplus B) - (X \ominus B)$

**Fig. 10.44** *Boundary detection of the input image X*

# Region Filling



- **Region filling** is the process of **filling** or “**colouring in**” **a defined area** of an image.
- A region can be defined at:
  - **Pixel level**
  - **Geometric level**
- **Types of Regions:**
  - **Boundary-defined region**
    - The region is defined by the **pixels forming its boundary**.
    - Algorithms used to fill these regions are called **Boundary-Fill Algorithms.**
  - **Interior-defined region**
    - The region is defined by its **interior pixels.**
    - Algorithms used are called **Flood-Fill Algorithms.**
- At the **geometric level**, regions are enclosed by elements such as **connected lines and curves.**

# Region Filling



| Boundary Fill | Flood Fill |
| --- | --- |
| Starts with a boundary | Starts with an interior seed |
| Boundary is known | Interior region is identified |
| Fills toward the inside | Expands from the inside |
| Stops at boundary | Stops when boundary is reached |

## Mathematical Representation

- $X_k = (X_{k-1} \oplus B) \cap A^c$

- **B** → Structuring element.
- **A** → Set containing the 8-connected boundary points of the region.
- **k** → Number of iterations.
- $A^c$ → Complement of the boundary.

# Region Filling



## How Region Filling Works

- Start with a point inside the boundary.
- Apply dilation using the structuring element B.
- The dilation expands the filled region.
- Take the intersection with the complement of the boundary to prevent crossing the boundary.
- Repeat the process iteratively.
- Continue until the entire region is filled with 1s.

## Termination Conditions

- The algorithm can be stopped in two ways:
- **Condition 1:** Stop when the entire region is filled.
- **Condition 2:** Predetermine the number of iterations required.

# Question



**Example 10.5** *The input image and structuring element are shown in Fig.* 10.46. *Perform the region filling operation.*

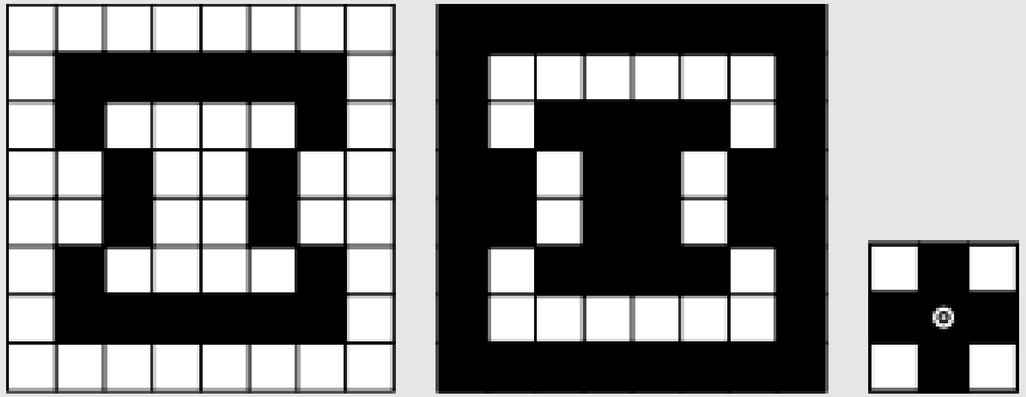
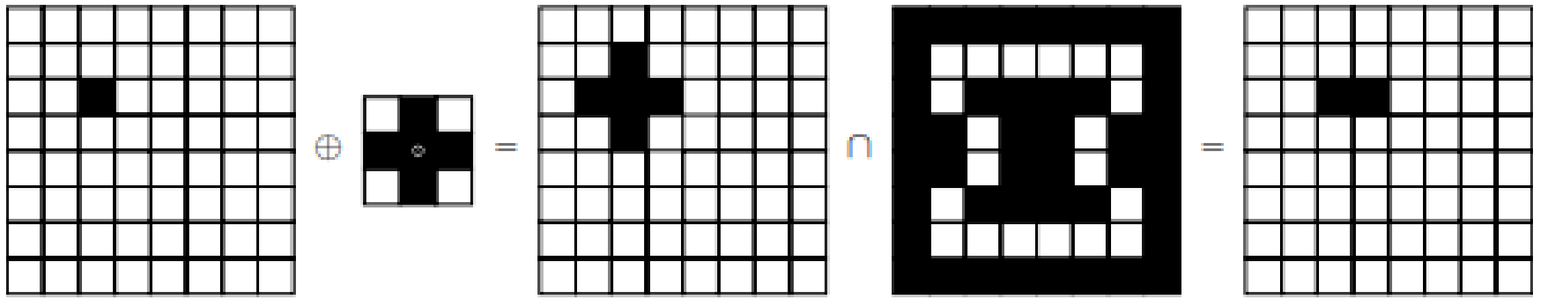



**Fig. 10.46** *(a) Input image A (b) Complement of input image (c) Structuring element B*

# Solution

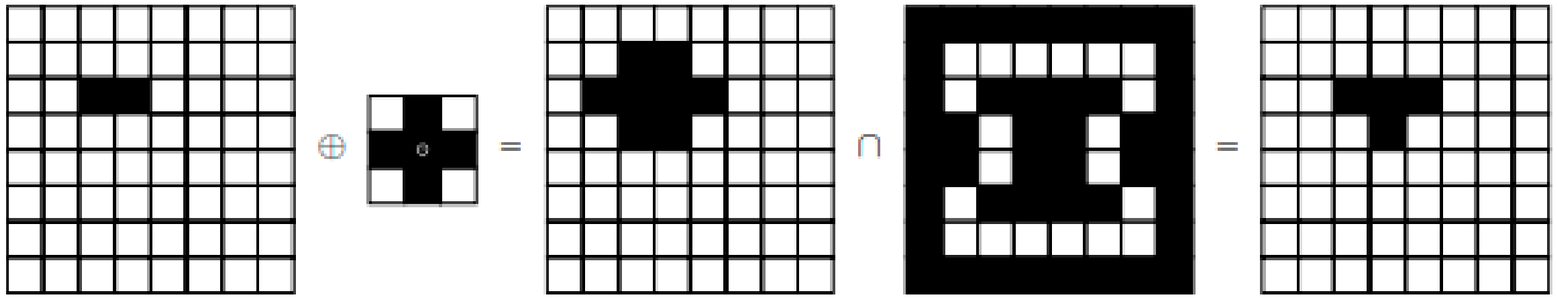


***Step 1*** Initially, take the $X_0$ as shown below. Now, perform the dilation of $X_0$ with the structuring element $B$. The resultant image is then intersected with the complement of the input image. This completes the first iteration.



$(X_0)$ $(B)$

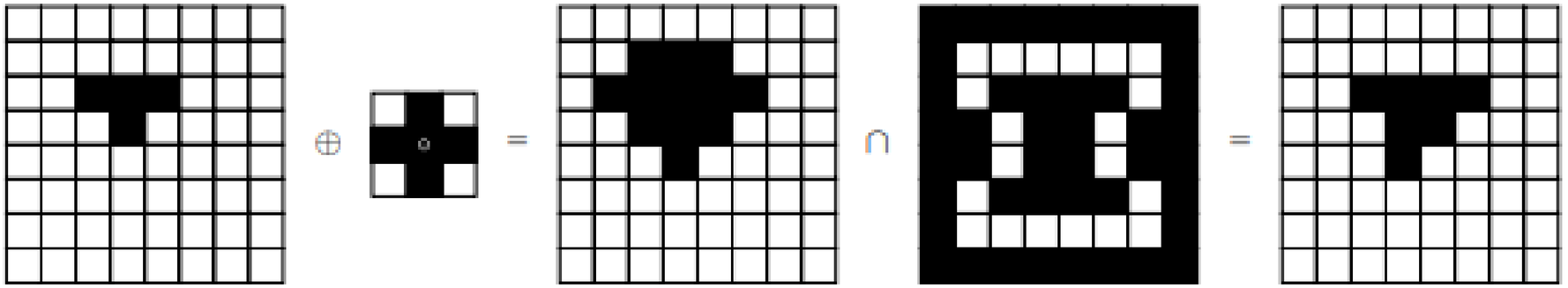
***Step 2*** Now, the input to the second step is the result of the first iteration. The process performed in Step 1 is repeated in Step 2.



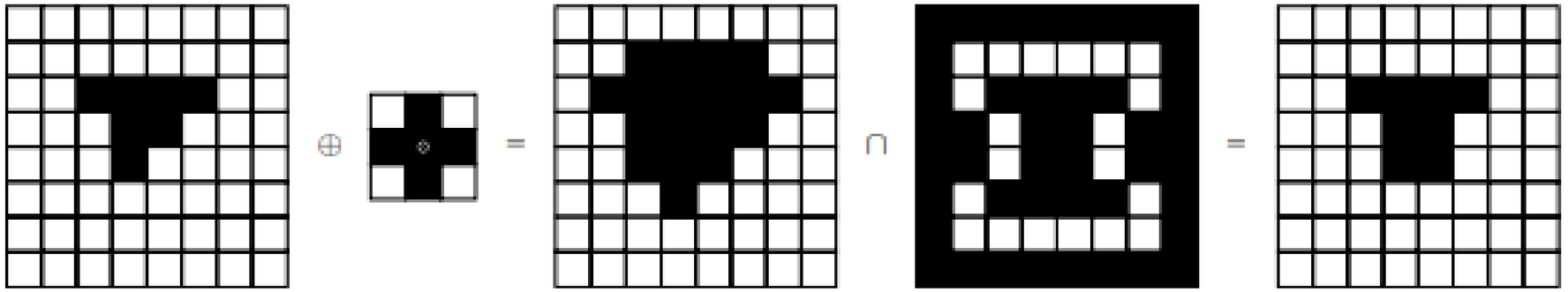
# Solution



***Step 3*** The same process is repeated again but the input image to the third iteration is the output image of Step 2.



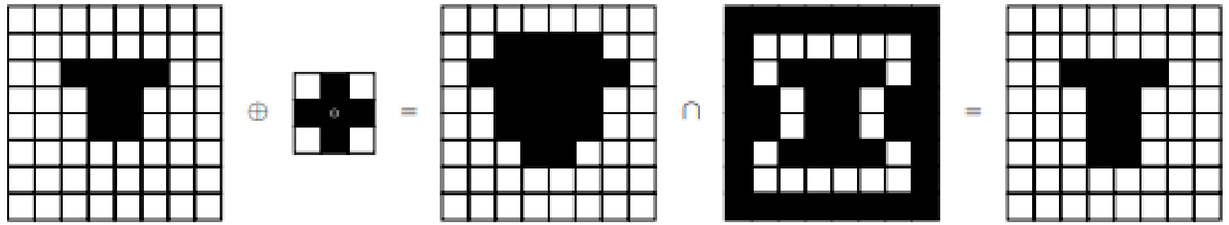
***Step 4*** The steps followed in steps 2 and 3 are repeated in Step 4. The resultant image is shown below.



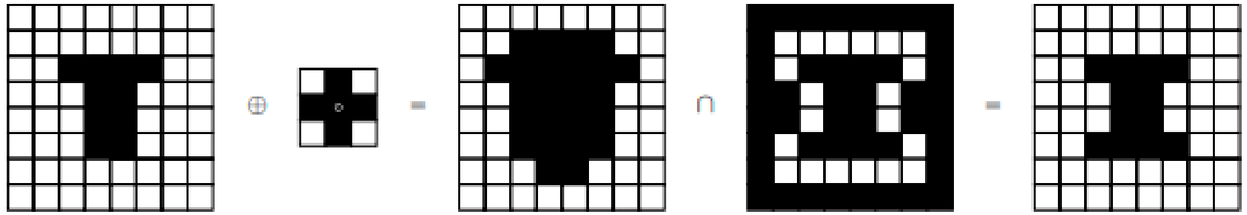
# Solution



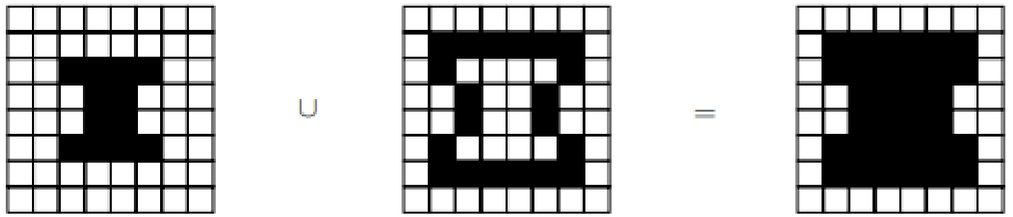
***Step 5*** The input to Step 5 is the output image of Step 4. The process done in Step 4 is repeated in Step 5.



***Step 6*** The input to Step 6 is the output image of Step 5. The process done in Step 5 is repeated in Step 6.



***Step 7*** Now, we perform the union of the result obtained in Step 7 with the original input image to get the region-filled image which is shown below.



# Convex Hull



- Convex Hull is a morphological operation used to obtain the **smallest convex shape** that **completely contains a binary object.**
- A set **X is convex** if the straight-line segment joining **any two points in X** lies completely within X.

**Basic Idea**

- Imagine placing an **elastic band around the binary object**.
- The band follows the outermost parts of the object.
- Any **concave portions or inward gaps are bridged** by the band.
- The resulting shape becomes **convex** while still containing the original object.

# Convex Hull



## Morphological Approach

- The convex hull is obtained using the **Hit-or-Miss Transform (HM)**.
- Four different structuring elements are used:
  - $B^1$
  - $B^2$
  - $B^3$
  - $B^4$
- The transformation is applied **iteratively** until the result converges.

## Mathematical Representation

$$Y_k^i = HM(Y_{k-1}^i, B^i) \cup X$$

**where:**

- **$i = 1,2,3,4$ → four structuring elements.**
- **$k = 1,2,3,\ldots$ → iteration number.**
- **$X$ → original binary image.**
- **$B^i$ → structuring element.**
- **HM → Hit-or-Miss Transform.**

# Convex Hull



**Initial condition:**

- $Y_0^i = X$

The process continues until convergence.

**Final Convex Hull**

- After convergence, the result obtained using each structuring element is:
  - $D^i = Y^i_{conv}$
- The final convex hull is the union of all four results:
  - $C(X) = \bigcup_{i=1}^{4} D^i$

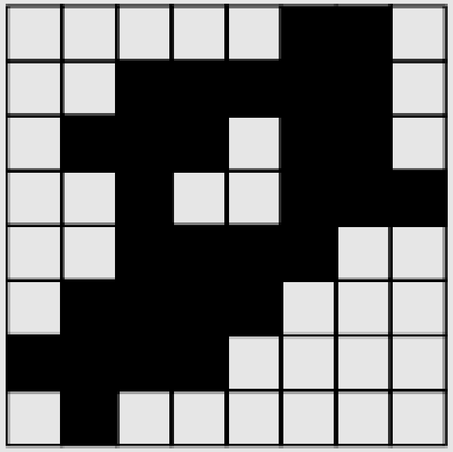
**Note:**

- **‘x’** in the structuring elements represents **don't care**.
- Four structuring elements are required to detect different orientations of convexity.
- Hit-or-Miss is repeatedly applied to identify and fill **concave** portions.
- The final result:
  - Has **no concavities.**
  - **Contains the original object**.
  - Represents the **convex hull** of the binary image.

# Question



**Example 10.6** *The input image X and the structuring element are shown below. Find the convex hull in the input image.*



$X = Y_0{}^1$

# Solution

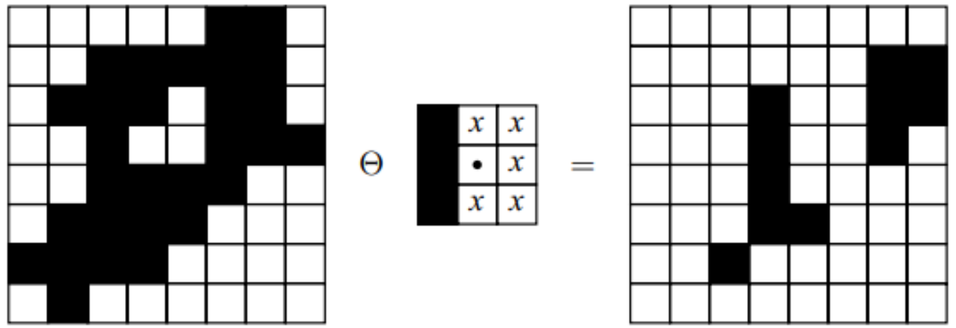


***Step 1*** The value of $Y_1^1$ is determined using $Y_1^1 = HM\left(Y_0^1, B^1\right) \cup X$.

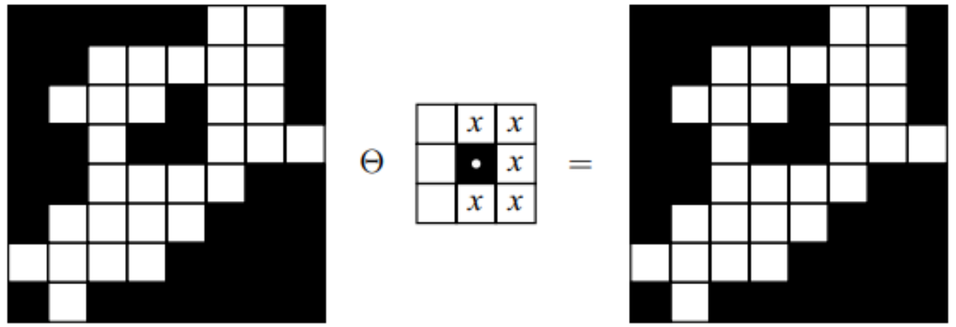
***Step 2*** To find $HM\left(Y_0^1, B^1\right)$

$$HM\left(Y_0^1, B^1\right) = \left(Y_0^1 \ominus B^1\right) \cap \left(\left(Y_0^1\right)^c \ominus \left(W - B^1\right)\right)$$

From the above definition, we have to find $Y_0^1 \ominus B^1$, the result of this operation is shown belo



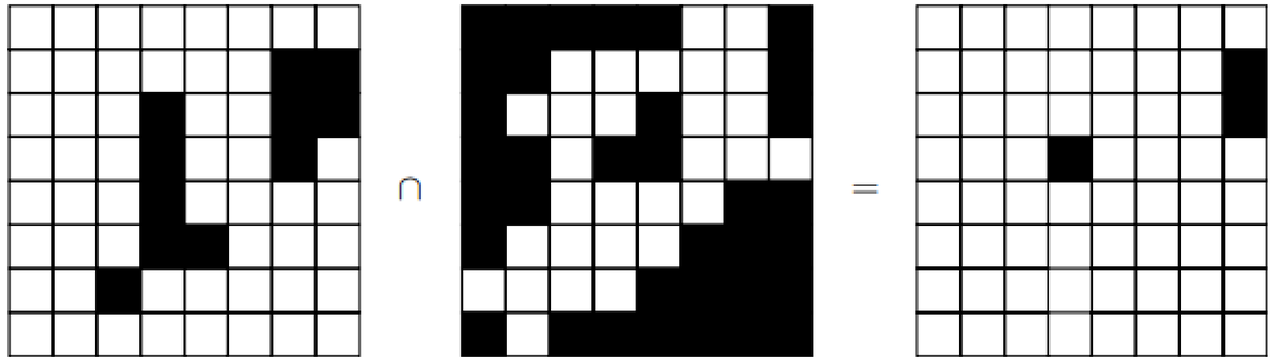

The next step is to find $\left(Y_0^1\right)^c \ominus \left(W - B^1\right)$. The result of $\left(Y_0^1\right)^c \ominus \left(W - B^1\right)$ is shown below.



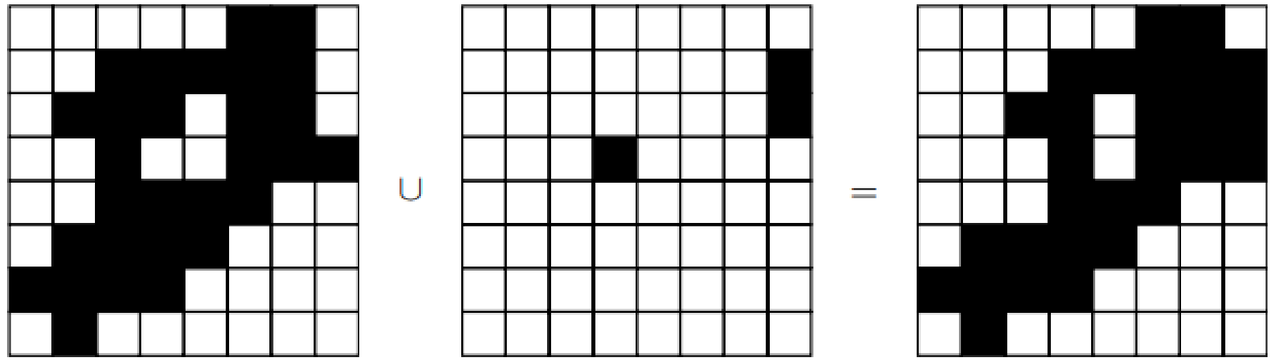
# Solution



Then we find the intersection of above two results which is illustrated below.



***Step 3*** The union of the input image with the result obtained in Step 2 will give the convex hull of the input image which is illustrated below.



# Thinning



- Thinning is the process of reducing a binary image to a unit-width skeleton.
- The skeleton represents the basic shape or structure of the object using only a thin set of pixels.
- It helps to:
  - Reduce the number of pixels.
  - Simplify the shape of an object.
  - Reduce the computational effort required for shape description and analysis.

# Thinning



## Morphological Basis

- Thinning is based on the Hit-or-Miss Transform.
- It can be represented as:
  - $X \otimes B = X - HM(X,B)$ **or**
  - $X \otimes B = X \cap [HM(X,B)]^c$
- HM(X,B) → Hit-or-Miss Transform.
- B → Structuring element.
- X → Input binary image.
- The operation removes selected boundary pixels while preserving the important structure of the object.

# Thinning



## Thinning Process – Type 1

- The process **removes boundary pixels** from a connected component.
- Pixels are removed only when they:
  - Are **not essential for maintaining connectivity.**
  - Do not represent important **geometrical features** of the object.
- The process is performed **iteratively**.
- It converges when:
  - The connected skeleton **does not change**, or
  - The skeleton **vanishes** even if iterations continue.

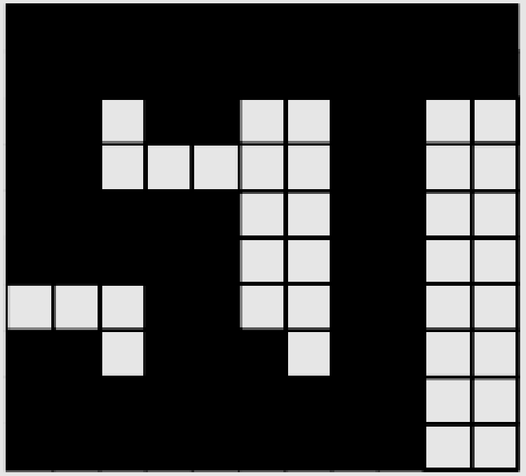
## Thinning Process – Type 2

- The process assigns a **distance label** to every pixel.
- The label represents the **distance of that pixel from the boundary.**
- Pixels with **local minimum distance labels** are identified.
- These pixels are then used to derive the **resulting skeleton**.

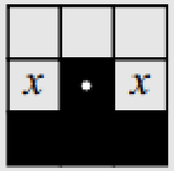
# Question



**Example 10.7** *Apply the thinning process to the image using the structuring element shown below.*



*Input image*



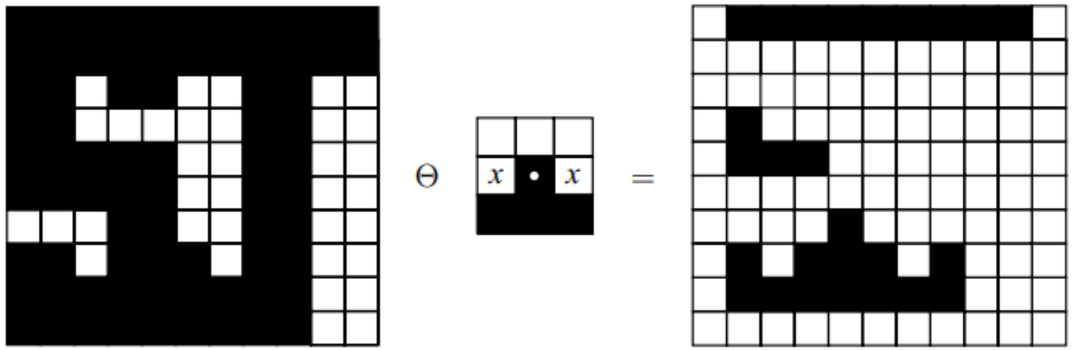

*Structuring element*

# Solution

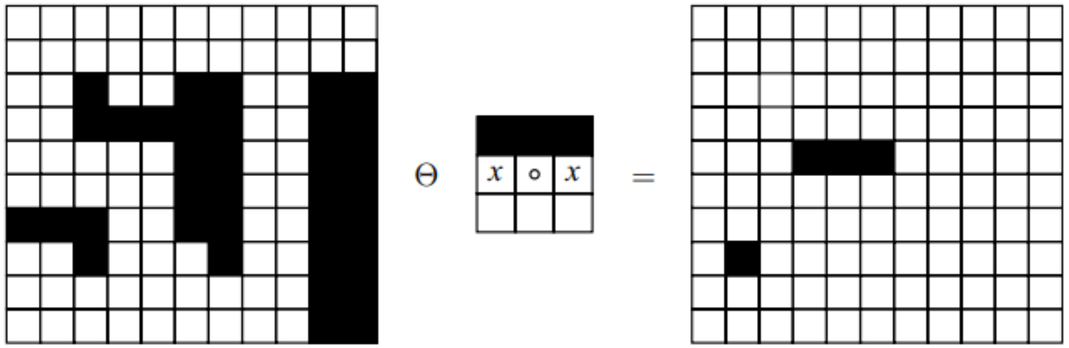


***Step 1*** *To perform the eroded operation of input image with structuring element*

First, we find the eroded input image with the structuring element. The resultant image is shown below.




***Step 2*** To perform the eroded operation of the complement input image with the complement structuring element. The resultant image is illustrated below.

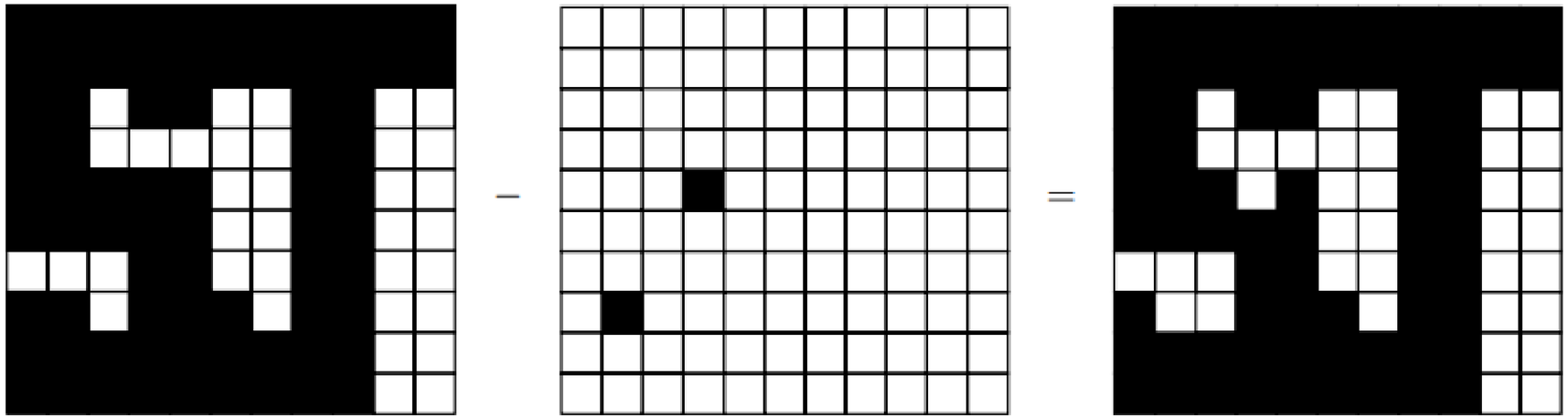


# Solution



***Step 3*** *To find the hit-or-miss transformation of the input image*

The hit-or-miss transformation is the intersection of the above two results. This resultant image is subtracted from the original input image; and we get the thinned original image. This is illustrated below.



**Note** Changing the structuring element will yield a different extent of the thinning operation.

# Thickening



- Thickening is a morphological operation used to grow selected regions of foreground pixels in a binary image.
- It is somewhat similar to dilation or closing.
- The operation adds new foreground pixels while preserving the original foreground pixels.

**Mathematical Representation**

$$X.B = X \cup HM(X,B)$$

**where:**

$X \rightarrow$ Input binary image.

$B \rightarrow$ Structuring element.

$HM(X,B) \rightarrow$ Hit-or-Miss Transform.

$\cup \rightarrow$ Union operation.

# Thickening



## How Thickening Works

- The thickened image consists of:
  - The original foreground pixels, plus
  - Additional pixels detected by the Hit-or-Miss Transform.
- The Hit-or-Miss Transform identifies locations where the structuring element matches the image.
- At those locations, the corresponding image pixel is changed to foreground.

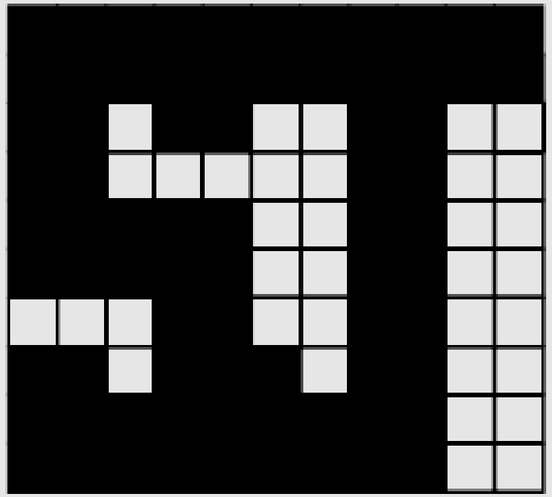
## Step-by-Step Process

- Place the structuring element B at each possible position in the image.
- Compare the black and white pattern of B with the corresponding image pixels.
- If the pattern exactly matches, the pixel under the origin of B is turned black (foreground).
- If there is no match, the pixel remains unchanged.
- Repeat the process using the required structuring elements.
- Continue iteratively until no further changes occur.

# Question



**Example 10.8** *Find the thickening process of the given image. Use $B_1$ as a structuring element which is shown in Fig.* 10.49.

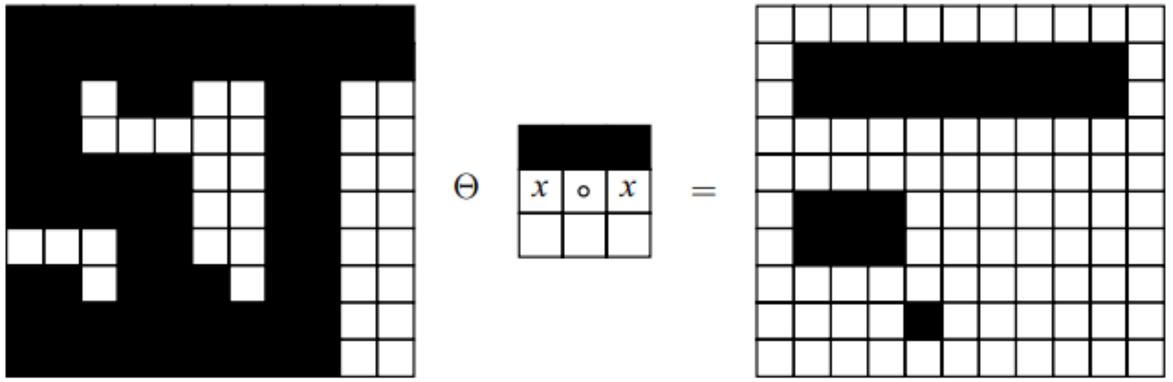


# Solution

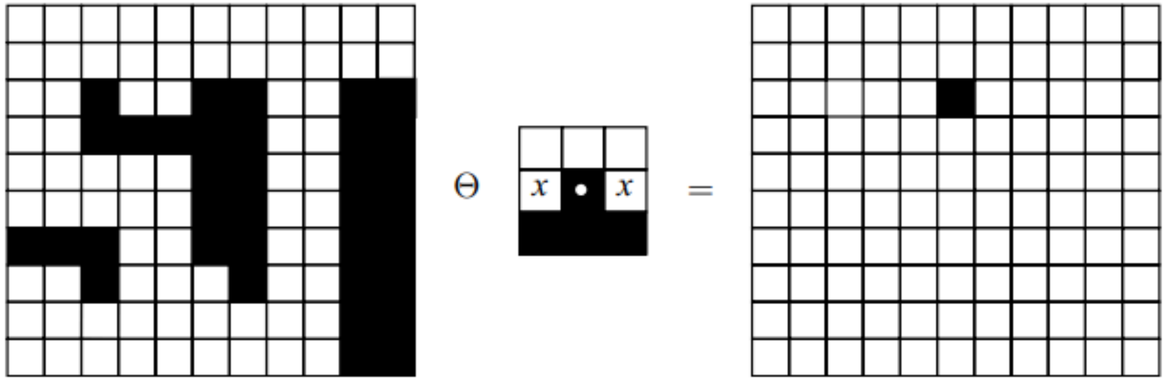


*Solution*

***Step 1*** First, find the hit-or-miss transformation of the given image with the structuring element $B_1$.



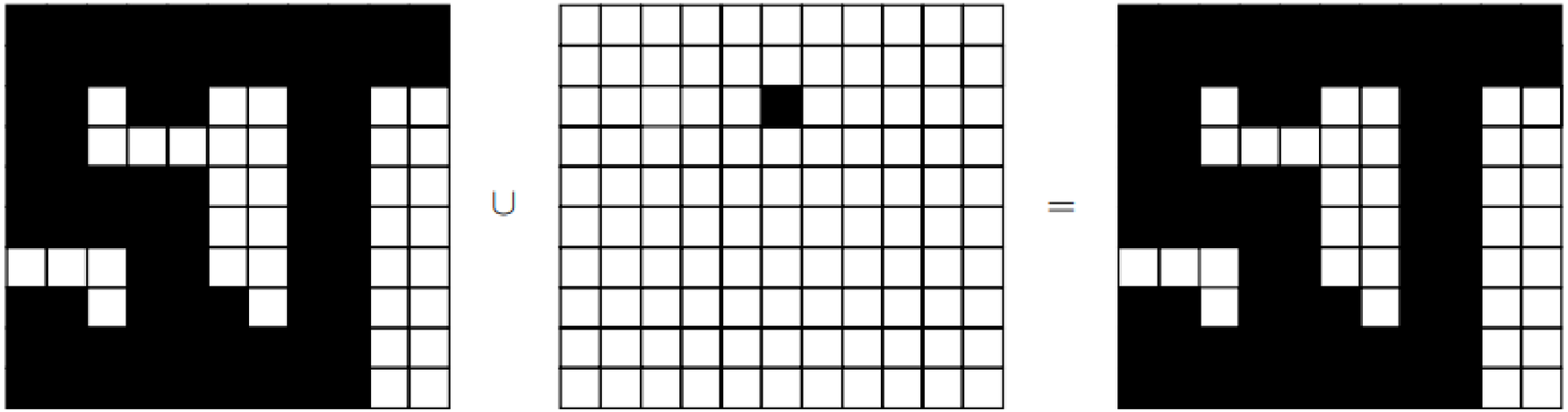

***Step 2*** Then find the erosion of the complement of the input image with the complement of the structuring element.



# Solution



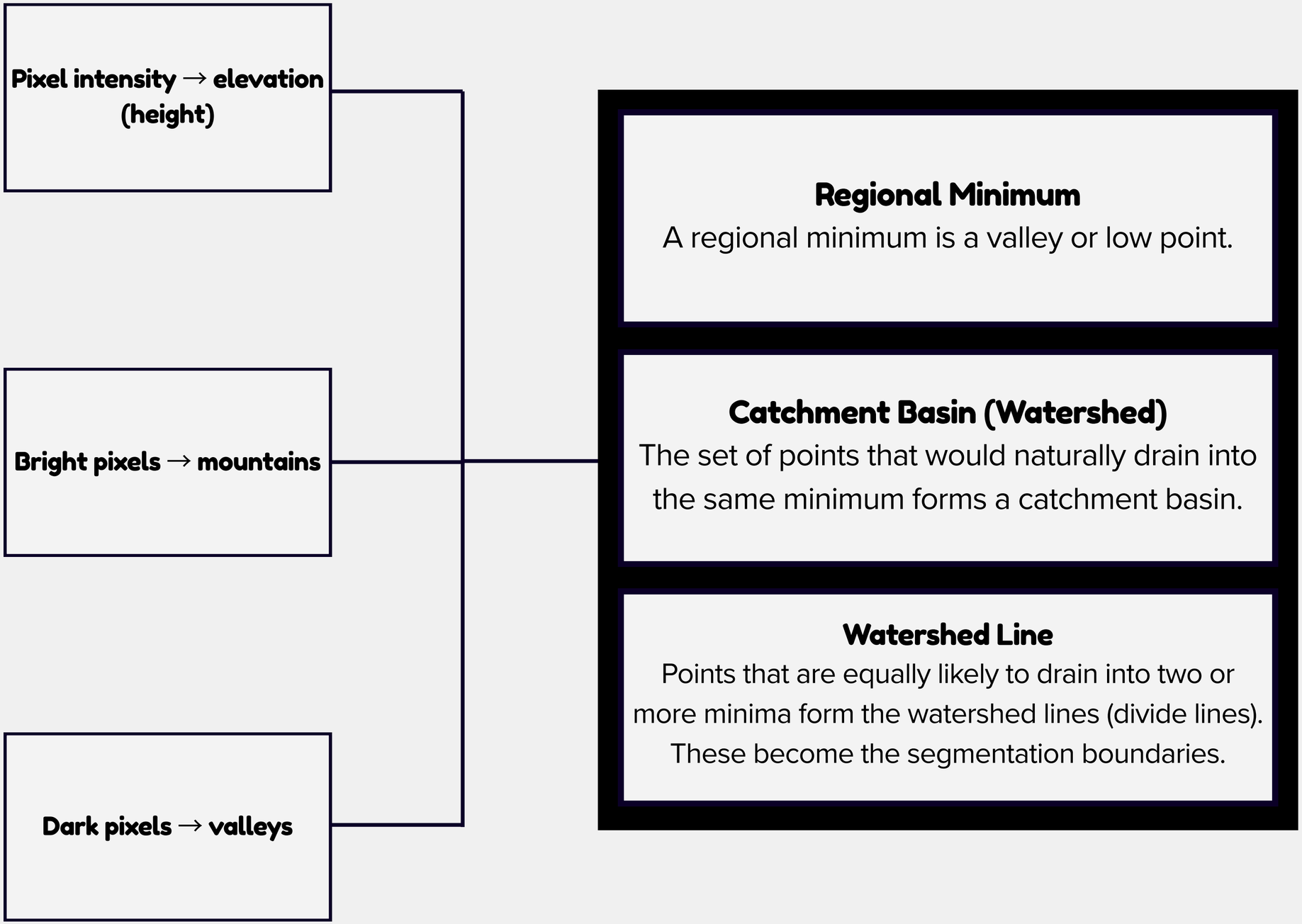
***Step 3*** Now, find the intersection of the output of steps 1 and 2. This result is combined (union) with the original image which is shown below.



# Summary

| Algorithm | Main Purpose | Simple Meaning | Applications |
|---|---|---|---|
| **Hit-or-Miss** | Detect a specific shape/pattern | "Find this exact pattern" | Pattern detection, shape recognition, feature detection |
| **Boundary Detection** | Extract the boundary of an object | "Give me the outline" | Object boundary extraction, shape analysis, segmentation |
| **Region Filling** | Fill an enclosed/closed region | "Fill the inside" | Filling holes, object segmentation, restoring missing regions |
| **Convex Hull** | Find the smallest convex shape around an object | "Put a rubber band around it" | Shape analysis, object recognition, defect/concavity detection |
| **Thinning** | Reduce objects to a thin skeleton | "Remove thickness, keep structure" | Character recognition, fingerprint analysis, skeletonization |
| **Thickening** | Increase the thickness of selected structures | "Make the object/line thicker" | Connecting broken structures, shape modification, feature enhancement |

# SEGMENTATION USING MORPHOLOGICAL WATERSHEDS


Pixel intensity → elevation (height)
Bright pixels → mountains
Dark pixels → valleys
Regional Minimum
A regional minimum is a valley or low point.
Catchment Basin (Watershed)
The set of points that would naturally drain into the same minimum forms a catchment basin.
Watershed Line
Points that are equally likely to drain into two or more minima form the watershed lines (divide lines). These become the segmentation boundaries.

# Flooding Analogy




1. A hole is drilled in every regional minimum.
2. Water rises from the bottom through these holes.
3. Water fills each basin gradually.
4. When water from two basins is about to merge, a dam is built.
5. Flooding continues until the highest level is reached.
6. The final dams become the watershed boundaries.

# Applying Watersheds to the Gradient Image

Watershed segmentation is often applied not directly to the image, but to its gradient image.

**Uniform regions → small gradient values**
**Object boundaries → large gradient values**

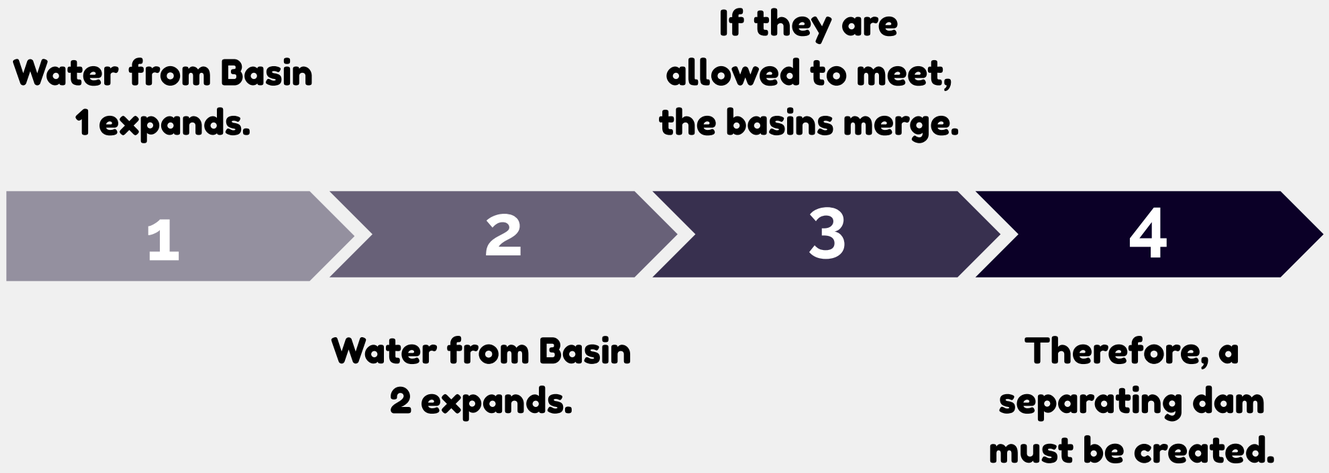
- **Regional minima in the gradient correspond to interiors of objects.**
- **High gradient ridges correspond to object edges.**

This makes watershed segmentation particularly effective for extracting blob-like objects from the background.

# Dam Construction Using Morphology

**The watershed method is called morphological watershed segmentation because it uses morphological operations, particularly dilation, to construct the dams.**

**When flooding reaches level n :**



# Morphological Dilation Step

Each flooded basin is dilated using a symmetric structuring element.

Two constraints are imposed:

1

### Condition 1

Dilation must stay inside the current connected component .

2

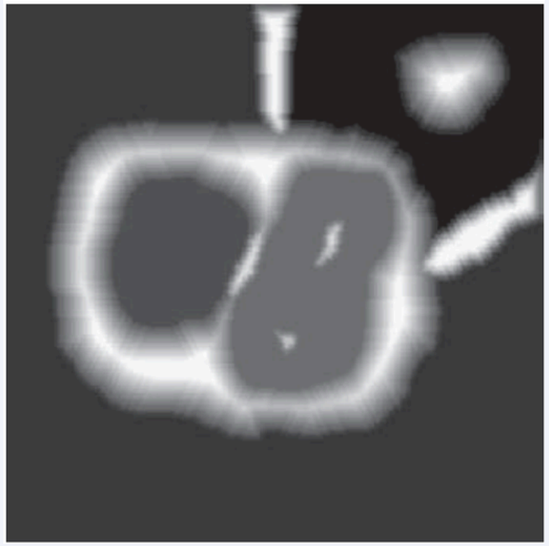
### Condition 2

Dilation must not allow two basins to merge.

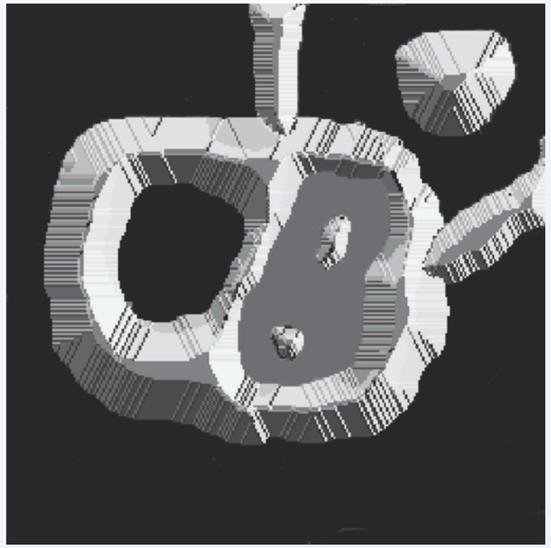
After repeated dilation:

- The only remaining pixels are those separating the basins.
- These pixels form a one-pixel-thick connected path.
- This path becomes the dam.

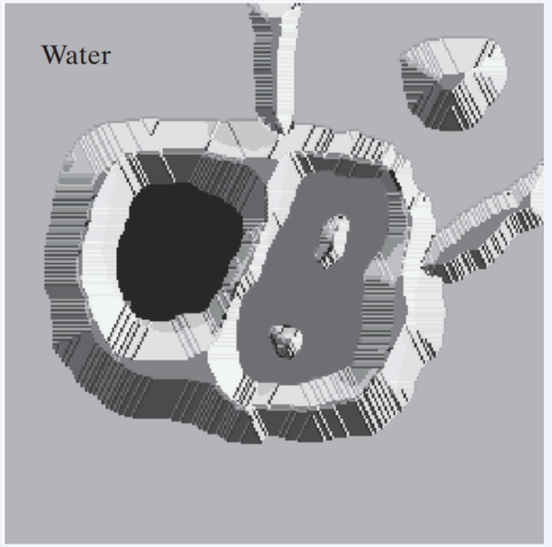
The dam pixels are assigned a value greater than the maximum image intensity so that water cannot cross them later.



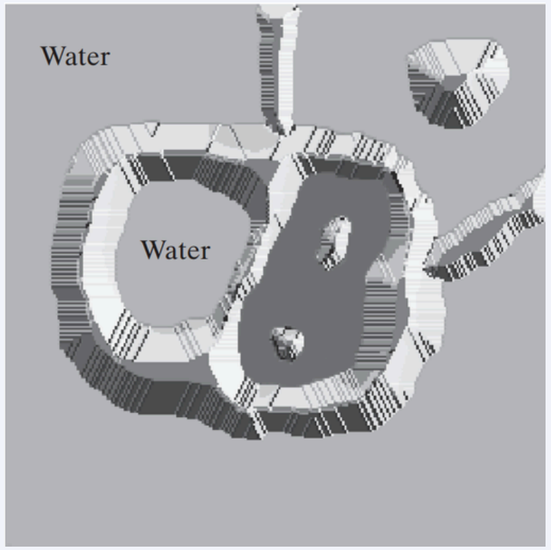
Original Grayscale image



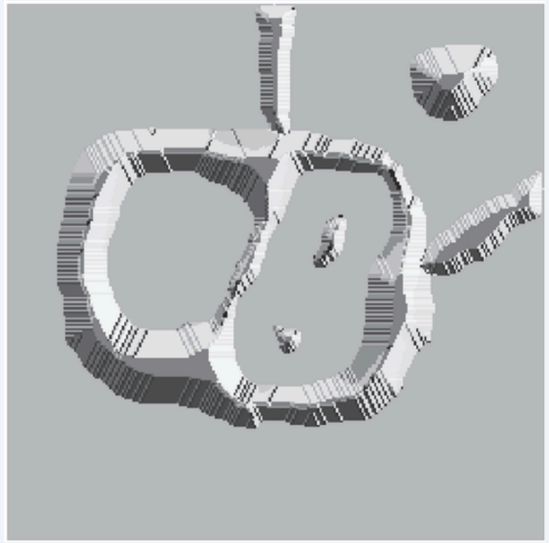
A topographic view represents an image as a 3D surface, where the height of each mountain corresponds to the pixel intensity. The backsides are shaded for easier viewing. The entire image is surrounded by dams higher than the maximum intensity, preventing water from flowing out of the image boundaries.



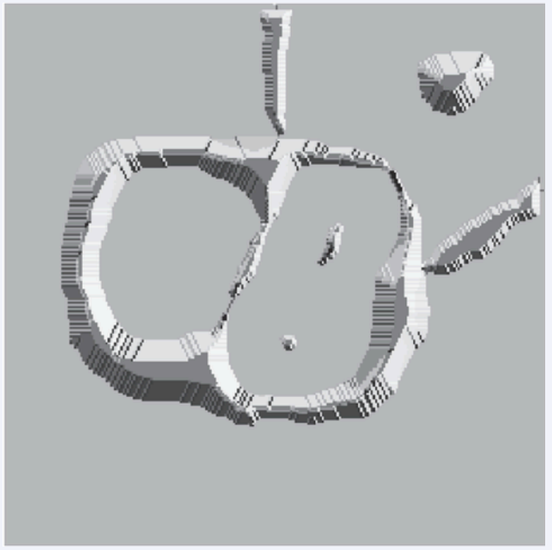

The image is flooded by raising water uniformly through holes in the darkest regions (regional minima). At the beginning, water fills only the lowest areas, which correspond to the black background of the image.



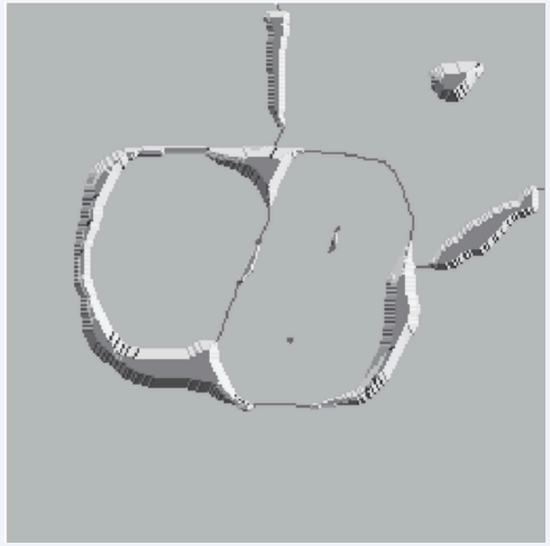

As the water level rises, it fills the first and second catchment basins.



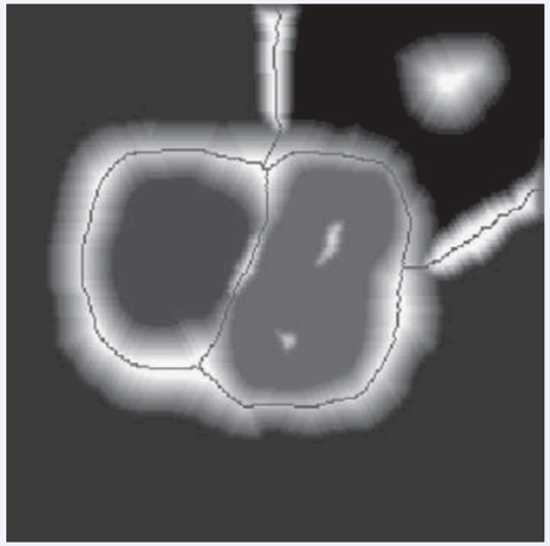
As the water level rises, it fills the first and second catchment basins.



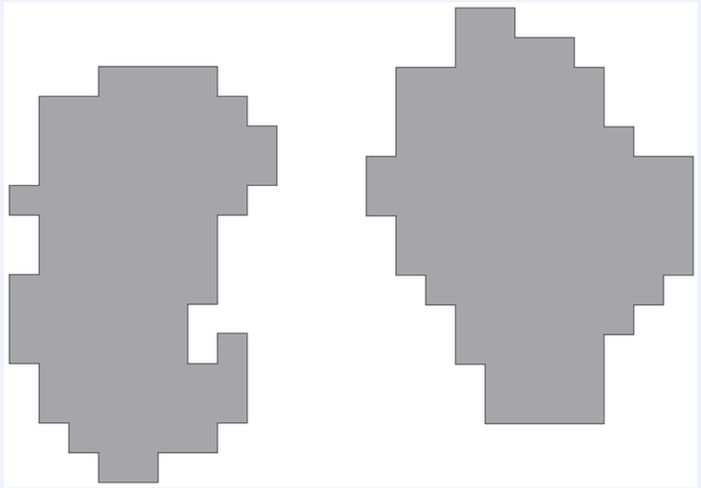
As the water level rises, water from one catchment basin begins to overflow into another. A small dam is built between the basins to stop the water from merging.



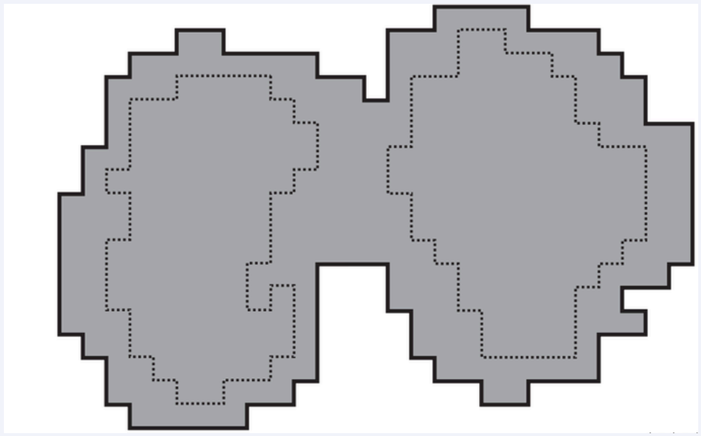
As flooding continues, longer dams are built between catchment basins to prevent their waters from merging. Another dam prevents water from mixing with the background region. This process continues until the maximum intensity level of the image is reached.



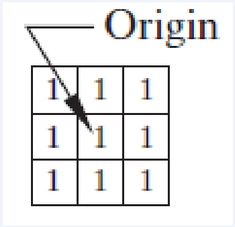
The final dams correspond to the watershed lines, which are the desired segmentation boundaries. The result for this example is shown. The dark, one-pixel-thick paths superimposed on the original image.



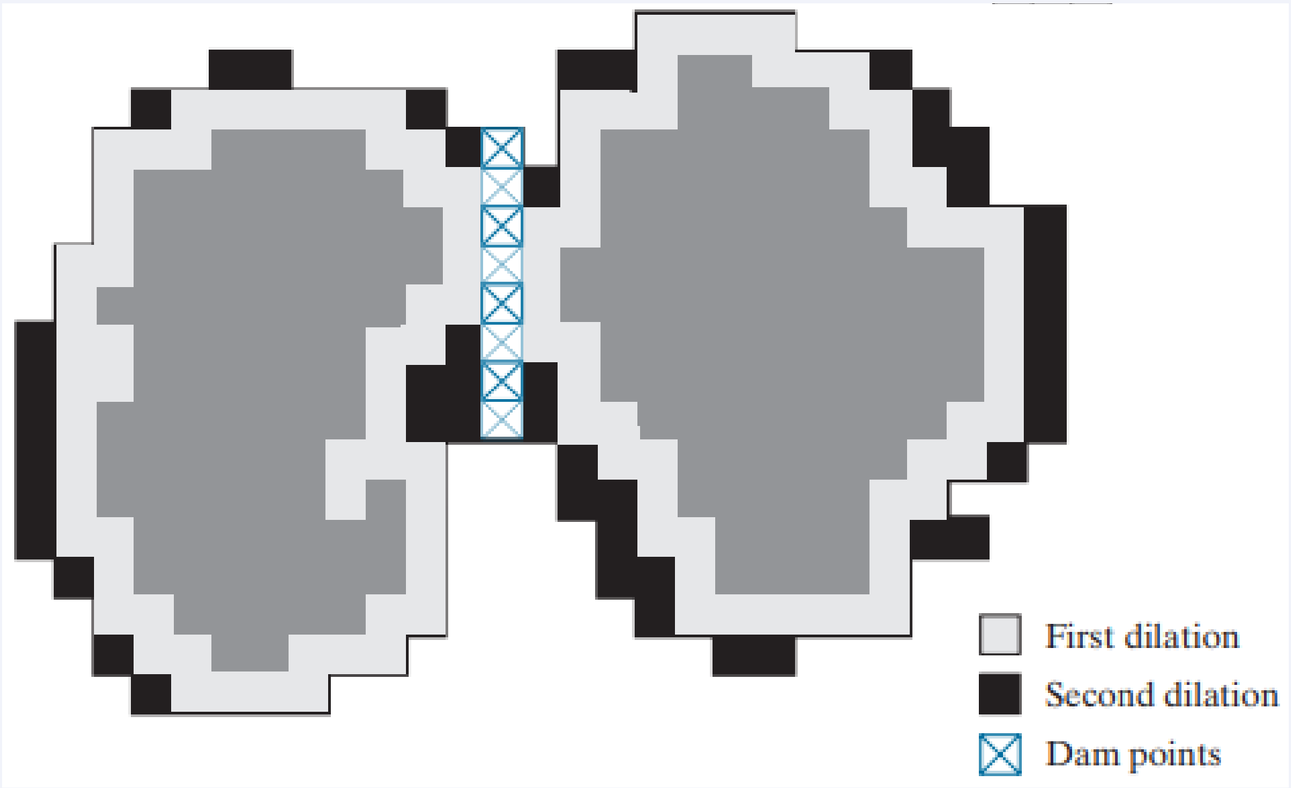
(a) Two partially flooded catchment basins at stage n – 1 of flooding.



(b) Flooding at stage n, showing that water has spilled between basins.




(c) Structuring element used for dilation.




(d) Result of dilation and dam construction.

# Thank You

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